

LIMIT

JEE EXERCISES

Exercise J-1

EVALUATION OF LIMITS USING STANDARD OPERATION

1. $\lim_{x \rightarrow 3} \frac{(x^3 + 27) \ln(x - 2)}{(x^2 - 9)} =$

- (A) -8 (B) 8 (C) 9 (D) -9

Ans. C

Sol. $\lim_{x \rightarrow 3} \frac{(x^3 + 27) \ln(x - 2)}{(x^2 - 9)} = \lim_{x \rightarrow 3} \frac{(x + 3)(x^2 - 3x + 9) \ln[1 + (x - 3)]}{(x - 3)(x + 3)} = \lim_{x \rightarrow 3} (x^2 - 3x + 9) \cdot \frac{\ln[1 + (x - 3)]}{(x - 3)}$
 $= (9 - 9 + 9)(1) = 9$

2. $\lim_{x \rightarrow 0} \frac{(4^x - 1)^3}{\sin\left(\frac{x}{p}\right) \ln\left(1 + \frac{x^2}{3}\right)} =$

- (A) $9p(\log 4)$ (B) $3p(\log 4)^3$ (C) $12p(\log 4)^3$ (D) $27p(\log 4)^2$

Ans. B

Sol. $\lim_{x \rightarrow 0} \frac{(4^x - 1)^3}{\sin\left(\frac{x}{p}\right) \ln\left(1 + \frac{x^2}{3}\right)} = \lim_{x \rightarrow 0} \frac{x^3 \left(\frac{4^x - 1}{x}\right)^3}{\left(\frac{x}{p}\right) \left[\frac{\sin\left(\frac{x}{p}\right)}{\frac{x}{p}}\right] \ln\left(1 + \frac{x^2}{3}\right)} = 3p \cdot \lim_{x \rightarrow 0} \frac{\left(\frac{4^x - 1}{x}\right)^3 \cdot \left(\frac{x^2}{3}\right)}{\left[\frac{\sin\left(\frac{x}{p}\right)}{\frac{x}{p}}\right] \ln\left(1 + \frac{x^2}{3}\right)} = 3p(\ln 4)^3$

3. $\lim_{x \rightarrow \infty} \frac{x^3 \cdot \sin \frac{1}{x} + x + 1}{x^2 + x + 1} =$

- (A) 0 (B) $1/2$ (C) 1 (D) none

Ans. C

Sol. $\lim_{x \rightarrow \infty} \frac{x^3 \sin \frac{1}{x} + x + 1}{x^2 + x + 1} = \lim_{x \rightarrow \infty} \frac{\frac{\sin \frac{1}{x}}{\frac{1}{x}} + \frac{1}{x} + \frac{1}{x^2}}{1 + \frac{1}{x} + \frac{1}{x^2}} = \frac{1 + 0 + 0}{1 + 0 + 0} = 1$

4. $\lim_{n \rightarrow \infty} \frac{-3n + (-1)^n}{4n - (-1)^n}$ is :

(A) $-\frac{3}{4}$

(B) $-\frac{3}{4}$ if n is even ; $\frac{3}{4}$ if n is odd

(C) not exist if n is even ; $-\frac{3}{4}$ if n is odd

(D) + 1 if n is even ; does not exist if n is odd

$$\lim_{n \rightarrow \infty} \frac{-3n + (-1)^n}{4n - (-1)^n} =$$

(A) $-\frac{3}{4}$

(B) $-\frac{3}{4}$ यदि n सम है; $\frac{3}{4}$ यदि n विषम है

(C) यदि n सम है तो विद्यमान नहीं है; $-\frac{3}{4}$ यदि n विषम है

(D) + 1 यदि n सम है; यदि n विषम है तो विद्यमान नहीं है

Ans. A

Sol. $\lim_{n \rightarrow \infty} \frac{-3n + (-1)^n}{4n - (-1)^n} = \lim_{n \rightarrow \infty} \frac{-3 + (-1)^n \cdot \frac{1}{n}}{4 - (-1)^n \cdot \frac{1}{n}} = \frac{-3 + 0}{4 - 0} = -\frac{3}{4}$

5. $\lim_{x \rightarrow 0^+} (1 + \tan^2 \sqrt{x})^{5/x} =$

(A) e^5 (B) e^2 (C) e (D) none

$\lim_{x \rightarrow 0^+} (1 + \tan^2 \sqrt{x})^{5/x} =$

(A) e^5 (B) e^2 (C) e (D) इनमें से कोई नहीं

Ans. A

Sol. $\lim_{x \rightarrow 0^+} (1 + \tan^2 \sqrt{x})^{5/x} = \lim_{h \rightarrow 0} (1 + \tan^2 \sqrt{0+h})^{5/(0+h)}$

$= \lim_{h \rightarrow 0} (1 + \tan^2 \sqrt{h})^{5/h} \quad (1^\infty \text{ form रूप})$

$= e^{\lim_{h \rightarrow 0} \frac{5 \tan^2 \sqrt{h}}{h}} = e^{\lim_{h \rightarrow 0} 5 \left(\frac{\tan \sqrt{h}}{\sqrt{h}} \right)^2} = e^5$

6. The limiting value of $(\cos x)^{1/\sin x}$ as $x \rightarrow 0$ is:

(A) 1 (B) e (C) 0 (D) None of these

यदि $x \rightarrow 0$ हो, तो $(\cos x)^{1/\sin x}$ का सीमान्त मान है :

- (A) 1 (B) e (C) 0 (D) इनमें से कोई नहीं

Ans. A

Sol. $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{\sin x}}$ (1^∞ form :i)

$$= e^{\lim_{x \rightarrow 0} \left(\frac{\cos x - 1}{\sin x} \right)} = e^{\lim_{x \rightarrow 0} \frac{-2 \sin^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}}} = e^{\lim_{x \rightarrow 0} -\tan \left(\frac{x}{2} \right)} = e^0 = 1$$

7. $\lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 2^{2n}}{5^n + 2^n + 3^{2n+3}} =$

- (A) 5 (B) 3 (C) 1 (D) zero

$$\lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 2^{2n}}{5^n + 2^n + 3^{2n+3}} =$$

- (A) 5 (B) 3 (C) 1 (D) शून्य

Ans. D

Sol. $\lim_{n \rightarrow \infty} \frac{5^{n+1} + 3^n - 2^{2n}}{5^n + 2^n + 3^{2n+3}} = \lim_{n \rightarrow \infty} \frac{5 \cdot 5^n + 3^n - 4^n}{5^n + 2^n + 27 \cdot 9^n} = \lim_{n \rightarrow \infty} \frac{5 \cdot \left(\frac{5}{9}\right)^n + \left(\frac{3}{9}\right)^n - \left(\frac{4}{9}\right)^n}{\left(\frac{5}{9}\right)^n + \left(\frac{2}{9}\right)^n + 27} = \frac{0+0-0}{0+0+27} = 0$

8. $\lim_{x \rightarrow \pi/2} \frac{\left(1 - \tan \frac{x}{2}\right)(1 - \sin x)}{\left(1 + \tan \frac{x}{2}\right)(\pi - 2x)^3}$ is :

- (A) 1/16 (B) -1/16 (C) 1/32 (D) -1/32

Ans. C

Sol. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\tan \left(\frac{\pi}{4} - \frac{x}{2} \right) (1 - \sin x)}{(\pi - 2x)^3}$

$$\text{Let } x = \frac{\pi}{2} + y : y \rightarrow 0 \Rightarrow \lim_{y \rightarrow 0} \frac{\tan \left(-\frac{y}{2} \right) (1 - \cos y)}{(-2y)^3}$$

$$= \lim_{y \rightarrow 0} \frac{-\tan \frac{y}{2} \cdot 2 \sin^2 \frac{y}{2}}{(-8)y^3} = \lim_{y \rightarrow 0} \frac{1}{32} \cdot \frac{\tan \frac{y}{2}}{\left(\frac{y}{2}\right)} \cdot \left[\frac{\sin \frac{y}{2}}{\frac{y}{2}} \right]^2 = \frac{1}{32}$$

9. If $f(x) = \sum_{\lambda=1}^n \left(x - \frac{1}{\lambda}\right) \left(x - \frac{1}{\lambda+1}\right)$ then $\lim_{n \rightarrow \infty} f(0)$ is :

- (A) 1 (B) -1 (C) 2 (D) None

यदि $f(x) = \sum_{\lambda=1}^n \left(x - \frac{1}{\lambda}\right) \left(x - \frac{1}{\lambda+1}\right)$ हो, तो $\lim_{n \rightarrow \infty} f(0)$ का मान है :

- (A) 1 (B) -1 (C) 2 (D) इनमें से कोई नहीं

Ans. A

Sol. We have,

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x$$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x$$

10. The value of $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x) \sin 5x}{x^2 \sin 3x}$ is :

$\lim_{x \rightarrow 0} \frac{(1 - \cos 2x) \sin 5x}{x^2 \sin 3x}$ का मान है :

- (A) 10/3 (B) 3/10 (C) 6/5 (D) 5/6

Ans. A

Sol. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x) \sin 5x}{x^2 \sin 3x} \left(\frac{0}{0} \text{ form} \right)$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} \cdot \frac{3x}{3x} \cdot \frac{5x}{5x} \cdot \frac{\sin 5x}{\sin 3x} = \frac{10}{3}$$

11. $\lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 3} \right)^x =$

- (A) e^4 (B) e^2 (C) e^3 (D) e

Ans. A

Sol. We have,

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \left(\frac{x^2 + 5x + 3}{x^2 + x + 2} \right)^x$$

$$= \lim_{x \rightarrow \infty} \left(1 + \frac{4x + 1}{x^2 + x + 2} \right)^x = e^{\lim_{x \rightarrow \infty} \frac{x(4x+1)}{x^2 + x + 2}} = e^4$$

12. If $\lim_{x \rightarrow 0} \frac{\log(3+x) - \log(3-x)}{x} = k$, the value of k is :

यदि $\lim_{x \rightarrow 0} \frac{\log(3+x) - \log(3-x)}{x} = k$, तो k का मान है :

- (A) $-\frac{2}{3}$ (B) 0 (C) $-\frac{1}{3}$ (D) $\frac{2}{3}$

Ans. D

Sol. $\lim_{x \rightarrow 0} \frac{\log(3+x) - \log(3-x)}{x} = k$

By L-Hospital's rule,

$$\lim_{x \rightarrow 0} \frac{\frac{1}{3+x} + \frac{1}{3-x}}{1} = k \Rightarrow \frac{2}{3} = k$$

13. Let $f(a) = g(a) = k$ and their n^{th} derivatives $f^n(a)$, $g^n(a)$ exist and are not equal for some n . Further if

$\lim_{x \rightarrow a} \frac{f(a)g(x) - f(x)g(a)}{g(x) - f(x)} = 4$ then the value of k is :

माना $f(a) = g(a) = k$ तथा उनके n वें अवकलज $f^n(a)$, $g^n(a)$ विद्यमान हो और वे n के किन्हीं मानों के लिए बराबर नहीं हों, यदि

$\lim_{x \rightarrow a} \frac{f(a)g(x) - f(x)g(a)}{g(x) - f(x)} = 4$ हो, तो k का मान है :

- (A) 0 (B) 4 (C) 2 (D) 1

Ans. B

Sol. We have, $\lim_{x \rightarrow a} \frac{l(a)g(x) - l(a) - g(a)l(x) + g(a)}{g(x) - l(x)} = 4$

$$\Rightarrow \lim_{x \rightarrow a} \frac{l(a)g'(x) - 0 - g(a)l'(x) + 0}{g'(x) - l'(x)} = 4$$

$$\Rightarrow \frac{l(a)g'(a) - g(a)l'(a)}{g'(a) - l'(a)} = 4$$

$$\Rightarrow \frac{k\{g'(a) - l'(a)\}}{\{g'(a) - l'(a)\}} = 4 \quad [\because f(a) = g(a) = k]$$

$$\Rightarrow k = 4$$

14. Let α & β be the distinct roots of $ax^2 + bx + c = 0$, then $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$ is equal to :

यदि α और β समीकरण $ax^2 + bx + c = 0$ के दो असमान मूल हैं, तो $\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$ का मान है :

(A) $\frac{a^2}{2} (\alpha - \beta)^2$ (B) 0 (C) $\frac{-a^2}{2} (\alpha - \beta)^2$ (D) $\frac{1}{2} (\alpha - \beta)^2$

Ans. A

Sol.
$$\lim_{x \rightarrow \alpha} \frac{1 - \cos(ax^2 + bx + c)}{(x - \alpha)^2}$$

$$= \lim_{x \rightarrow \alpha} \left(\frac{1 - \cos a(x - \alpha)(x - \beta)}{(x - \alpha)^2 (x - \beta)^2 a^2} \right) (x - \beta)^2 a^2$$

$$= \frac{1}{2} a^2 (\alpha - \beta)^2$$

15. $\lim_{x \rightarrow 0^+} \frac{\ln(-\cos 3x + \cos x)}{\ln \sin x}$ is equal to :

$\lim_{x \rightarrow 0^+} \frac{\ln(-\cos 3x + \cos x)}{\ln \sin x}$ का मान होगा :

- (A) 0 (B) 1
 (C) 2 (D) D.N.E.

Ans. C

Sol.
$$\lim_{x \rightarrow 0^+} \frac{\ln(-4 \cos^3 x + 3 \cos x + \cos x)}{\ln \sin x}$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(4 \cos x \sin^2 x)}{\ln \sin x}$$

$$\lim_{x \rightarrow 0^+} \frac{\ln(4 \cos x) + \ln \sin^2 x}{\ln \sin x}$$

$$= \lim_{x \rightarrow 0^+} \frac{2 \ln \sin x}{\ln \sin x} + \frac{\ln(4 \cos x)}{\ln \sin x} = 2 + 0 = 2$$

BASED ON SPECIAL SIGNS LIKE GIF, MOD, SGN ETC

16. The value of $\lim_{x \rightarrow 0} (\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2 \tan^{-1}[\tan x])$ is equal to : [Note : $[k]$ denotes largest integer function less than or equal to k .]

- (A) π (B) $\pi/2$ (C) $3\pi/2$ (D) Non-existent

$\lim_{x \rightarrow 0} (\sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2 \tan^{-1}[\tan x])$ का मान होगा : (जहाँ $[.]$ महत्तम पूर्णांक फलन दर्शाता है)

- (A) π (B) $\pi/2$ (C) $3\pi/2$ (D) विद्यमान नहीं होगा

Ans. B

Sol. $\lim_{x \rightarrow 0} \sin^{-1}[\sin x] + \cos^{-1}[\cos x] - 2 \tan^{-1}[\tan x]$

LHL

$$= \sin^{-1}(-1) + \cos^{-1}(0) - 2 \tan^{-1}(-1)$$

$$= -\frac{\pi}{2} + \frac{\pi}{2} + 2 \cdot \frac{\pi}{4} = \frac{\pi}{2}$$

$$\text{RHL} = \sin^{-1}(0) + \cos^{-1}(0) - 2 \tan^{-1}(0) = \frac{\pi}{2}$$

17. Given $l_1 = \lim_{x \rightarrow \frac{\pi}{4}} \cos^{-1} \left[\sec \left(x - \frac{\pi}{4} \right) \right]; l_2 = \lim_{x \rightarrow \frac{\pi}{4}} \sin^{-1} \left[\operatorname{cosec} \left(x + \frac{\pi}{4} \right) \right];$

$$l_3 = \lim_{x \rightarrow \frac{\pi}{4}} \tan^{-1} \left[\cot \left(x + \frac{\pi}{4} \right) \right]; l_4 = \lim_{x \rightarrow \frac{\pi}{4}} \cot^{-1} \left[\tan \left(x - \frac{\pi}{4} \right) \right]$$

where $[x]$ denotes greatest integer function then which of the following limits exist?

(A) l_1 and l_2 only (B) l_1 and l_3 only (C) l_1 and l_4 only (D) All of them

दिया गया है $l_1 = \lim_{x \rightarrow \frac{\pi}{4}} \cos^{-1} \left[\sec \left(x - \frac{\pi}{4} \right) \right]; l_2 = \lim_{x \rightarrow \frac{\pi}{4}} \sin^{-1} \left[\operatorname{cosec} \left(x + \frac{\pi}{4} \right) \right];$

$$l_3 = \lim_{x \rightarrow \frac{\pi}{4}} \tan^{-1} \left[\cot \left(x + \frac{\pi}{4} \right) \right]; l_4 = \lim_{x \rightarrow \frac{\pi}{4}} \cot^{-1} \left[\tan \left(x - \frac{\pi}{4} \right) \right]$$

(जहाँ $[.]$ महत्तम पूर्णांक फलन दर्शाता है) तो निम्न में से कौनसी सीमा विद्यमान होगी

(A) केवल l_1 तथा l_2 (B) केवल l_1 तथा l_3 (C) केवल l_1 तथा l_4 (D) सभी

Ans. A

Sol. $l_1 = \lim_{x \rightarrow \frac{\pi}{4}} \cos^{-1} \left[\sec \left(x - \frac{\pi}{4} \right) \right]$

$$\text{LHL} = \lim_{h \rightarrow 0} \cos^{-1} [\sec(-h)] = 0$$

$$\text{RHL} = \lim_{h \rightarrow 0} \cos^{-1} [\sec h] = 0$$

$$l_2 = \lim_{x \rightarrow \frac{\pi}{4}} \sin^{-1} \left[\operatorname{cosec} \left(x + \frac{\pi}{4} \right) \right]$$

$$\text{LHL} = \lim_{h \rightarrow 0} \sin^{-1} \left[\operatorname{cosec} \left(\frac{\pi}{2} - h \right) \right] = \frac{\pi}{2}$$

$$\text{RHL} = \lim_{h \rightarrow 0} \sin^{-1} \left[\cos \sec \left(\frac{\pi}{2} + h \right) \right]$$

$$= \lim_{h \rightarrow 0} \sin^{-1} [\sec(h)] = \frac{\pi}{2}$$

$$l_3 = \lim_{x \rightarrow \frac{\pi}{4}} \tan^{-1} \left[\cot \left(x + \frac{\pi}{4} \right) \right]$$

$$\text{LHL} = \lim_{h \rightarrow 0} \tan^{-1} \left[\cot \left(\frac{\pi}{2} - h \right) \right] = \lim_{h \rightarrow 0} \tan^{-1} [\tanh] = 0$$

$$\text{RHL} = \lim_{h \rightarrow 0} \tan^{-1} \left[\cot \left(\frac{\pi}{2} + h \right) \right] = \lim_{h \rightarrow 0} \tan^{-1} [-\tanh] = -\frac{\pi}{4}$$

$\Rightarrow l_3$ does not exist

$$l_4 = \lim_{x \rightarrow \frac{\pi}{4}} \cot^{-1} \left[\tan \left(x - \frac{\pi}{4} \right) \right]$$

$$\text{LHL} = \lim_{h \rightarrow 0} \cot^{-1} [-\tanh] = -\frac{\pi}{4}$$

$$\text{RHL} = \lim_{h \rightarrow 0} \cot^{-1} [\tanh] = \cot^{-1}(0) = \frac{\pi}{2}$$

$\therefore l_4$ does not exist.

18. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{\cos^{-1} \left[\frac{1}{4} (3 \sin x - \sin 3x) \right]}$ where $[]$ denotes greatest integer function, is :

- (A) $2/\pi$ (B) 1 (C) $4/\pi$ (D) Does not exist

$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{\cos^{-1} \left[\frac{1}{4} (3 \sin x - \sin 3x) \right]}$ का मान होगा (जहाँ $[.]$ महत्तम पूर्णांक फलन दर्शाता है) :

- (A) $2/\pi$ (B) 1 (C) $4/\pi$ (D) विद्यमान नहीं है

Ans. A

Sol. $\lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{\cos^{-1} \left[\frac{1}{4} (3 \sin x - \sin 3x) \right]}$

$$= \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{\cos^{-1}[\sin^3 x]} = \frac{1}{\cos^{-1}(0)} = \frac{2}{\pi}$$

19. The value of $\lim_{x \rightarrow 0} \frac{(\tan(\{x\}-1)) \sin \{x\}}{\{x\}(\{x\}-1)}$ where $\{x\}$ denotes the fractional part function :

(A) is 1 (B) is $\tan 1$ (C) is $\sin 1$ (D) is non existent

$$\lim_{x \rightarrow 0} \frac{(\tan(\{x\}-1)) \sin \{x\}}{\{x\}(\{x\}-1)} \text{ का मान होगा (जहाँ } \{.\} \text{ भिन्नात्मक भाग दर्शाता है)}$$

(A) is 1 (B) is $\tan 1$ (C) is $\sin 1$ (D) विद्यमान नहीं है

Ans. D

Sol. $\lim_{x \rightarrow 0} \frac{(\tan(\{x\}-1)) \sin \{x\}}{\{x\}(\{x\}-1)} = \lim_{x \rightarrow 0^+} f(x) = \lim_{h \rightarrow 0^+} \frac{\tan(h-1) \cdot \sinh}{h(h-1)} = \frac{\tan(-1)}{-1} = \tan 1$

$$\lim_{x \rightarrow 0^-} \frac{\tan((1-h)-1) \sin(1-h)}{(1-h)(1-h-1)} = \frac{\sin 1}{1} = \sin 1$$

Hence $\lim_{x \rightarrow 0} f(x)$ does not exist]

20. $\lim_{x \rightarrow 1} (1-x + [x-1] + [1-x]) =$ where $[x]$ denotes greatest integer function :

(A) 0 (B) 1 (C) -1 (D) does not exist

$\lim_{x \rightarrow 1} (1-x + [x-1] + [1-x])$, जहाँ $[x]$ महत्तम पूर्णांक फलन है, का मान है :

(A) 0 (B) 1 (C) -1 (D) विद्यमान नहीं है ।

Ans. C

Sol. $\lim_{x \rightarrow 1} (1-x + [x-1] + [1-x]) = -1$

as

$$\text{LHL} = \lim_{h \rightarrow 0} (1 - (1-h) + [-h] + [1-1+h]) = -1$$

$$\text{RHL} = \lim_{h \rightarrow 0} (1 - (1+h) + [1+h-1] + [1-(1+h)]) = -1$$

21. The value of $\lim_{x \rightarrow \pi/4} (1+[x])^{1/\ln(\tan x)}$ (where $[.]$ denotes the greatest integer function) is equal to :

$\lim_{x \rightarrow \pi/4} (1+[x])^{1/\ln(\tan x)}$, जहाँ $[.]$ महत्तम पूर्णांक फलन है, का मान है :

(A) 0 (B) 1 (C) e (D) e^{-1}

Ans. B

Sol. $\lim_{x \rightarrow \frac{\pi}{4}} (1 + [x])^{\frac{1}{\ell n(\tan x)}} = \lim_{x \rightarrow \frac{\pi}{4}} (\text{exact } 1)^{\frac{1}{\ell n(\tan x)}} = 1$

22. The limit $\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right)$, where $[x]$ is the greatest integer function and $n \in \mathbb{N}$, is :

- (A) $2n$ (B) $2n + 1$ (C) $2n - 1$ (D) does not exist

$\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right)$, जहाँ $[x]$ महत्तम पूर्णांक फलन है और $n \in \mathbb{I}$, है :

- (A) $2n$ (B) $2n + 1$ (C) $2n - 1$ (D) विद्यमान नहीं है

Ans. C

Sol. $\sin \theta < \theta < \tan \theta$, $\theta \in \left(0, \frac{\pi}{2} \right)$

$$\frac{\sin \theta}{\theta} < 1 < \frac{\tan \theta}{\theta}$$

$$\frac{n \sin \theta}{\theta} < n < \frac{n \tan \theta}{\theta}$$

$$\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right); \quad n \in \mathbb{N}$$

$$\text{L.H.L.} = \lim_{\theta \rightarrow 0^-} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right) = n - 1 + n = 2n - 1$$

$$\text{R.H.L.} = \lim_{\theta \rightarrow 0^+} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right) = n - 1 + n = 2n - 1$$

$$\therefore \text{L.H.L.} = \text{R.H.L.} = 2n - 1$$

Hindi. $\sin \theta < \theta < \tan \theta$, $\theta \in \left(0, \frac{\pi}{2} \right)$

$$\frac{\sin \theta}{\theta} < 1 < \frac{\tan \theta}{\theta}$$

$$\frac{n \sin \theta}{\theta} < n < \frac{n \tan \theta}{\theta}$$

$$\lim_{\theta \rightarrow 0} \left(\left[\frac{n \sin \theta}{\theta} \right] + \left[\frac{n \tan \theta}{\theta} \right] \right); \quad n \in \mathbb{N}$$

$$\text{बाँयीं सीमा} = \lim_{\theta \rightarrow 0^-} \left(\left\lfloor \frac{n \sin \theta}{\theta} \right\rfloor + \left\lfloor \frac{n \tan \theta}{\theta} \right\rfloor \right) = n - 1 + n = 2n - 1$$

$$\text{दाँयीं सीमा} = \lim_{\theta \rightarrow 0^+} \left(\left\lfloor \frac{n \sin \theta}{\theta} \right\rfloor + \left\lfloor \frac{n \tan \theta}{\theta} \right\rfloor \right) = n - 1 + n = 2n - 1$$

$$\text{बाँयीं सीमा} = \text{दाँयीं सीमा} = 2n - 1$$

SANDWICH THEOREM

23. The value of $\lim_{x \rightarrow 0} (x^{16} + 4x^{12} - 3x^4) \left\lfloor \frac{1}{x^4} \right\rfloor$, (where $\lfloor \cdot \rfloor$ denotes greatest integer function), is :

- (A) 0 (B) -3 (C) 4 (D) Non-existent

$\lim_{x \rightarrow 0} (x^{16} + 4x^{12} - 3x^4) \left\lfloor \frac{1}{x^4} \right\rfloor$ का मान होगा, (जहाँ $\lfloor \cdot \rfloor$ महत्तम पूर्णांक फलन दर्शाता है) :

- (A) 0 (B) -3 (C) 4 (D) विद्यमान नहीं होगी

Ans. B

Sol. $\frac{1}{x^4} - 1 < \left\lfloor \frac{1}{x^4} \right\rfloor \leq \frac{1}{x^4}$

$$\Rightarrow (x^{16} + 4x^{12} - 3x^4) \frac{1 - x^4}{x^4} < (x^{16} + 4x^{12} - 3x^4) \left\lfloor \frac{1}{x^4} \right\rfloor \leq \frac{x^{16} + 4x^{12} - 3x^4}{x^4}$$

$$\Rightarrow \lim_{x \rightarrow 0} (x^{16} + 4x^{12} - 3x^4) \left(\frac{1 - x^4}{x^4} \right) < \lim_{x \rightarrow 0} (x^{16} + 4x^{12} - 3x^4) \left\lfloor \frac{1}{x^4} \right\rfloor \leq \lim_{x \rightarrow 0} \frac{x^{16} + 4x^{12} - 3x^4}{x^4}$$

$$\Rightarrow -3 < \lim_{x \rightarrow 0} (x^{16} + 4x^{12} - 3x^4) \left(\frac{1}{x^4} \right) \leq -3$$

$\therefore$ By sandwich theorem

$$\lim_{x \rightarrow 0} (x^{16} + 4x^{12} - 3x^4) \left(\frac{1}{x^4} \right) = -3$$

Aliter : $\lim_{x \rightarrow 0} (x^{16} + 4x^{12} - 3x^4) \left(\frac{1}{x^4} - \left\{ \frac{1}{x^4} \right\} \right) = -3$

$$\because 0 \leq \left\{ \frac{1}{x^4} \right\} < 1$$

therefore limit = -3

24. If $[x]$ denotes the greatest integer $\leq x$, then $\lim_{n \rightarrow \infty} \frac{1}{n^4} ([1^3 x] + [2^3 x] + \dots + [n^3 x])$ equals :

यदि $[x]$, x से छोटे एवं बराबर महत्तम पूर्णांक को प्रदर्शित करता है, तो $\lim_{n \rightarrow \infty} \frac{1}{n^4} ([1^3 x] + [2^3 x] + \dots + [n^3 x])$ का मान है :

- (A) $x/2$ (B) $x/3$ (C) $x/6$ (D) $x/4$

Ans. D

Sol. $\lim_{n \rightarrow \infty} \frac{1}{n^4} ([1^3 x] + [2^3 x] + \dots + [n^3 x])$

$$1^3 x - 1 < [1^3 x] \leq 1^3 x$$

$$2^3 x - 1 < [2^3 x] \leq 2^3 x$$

$$\cdot \quad \cdot \quad \cdot$$

$$\cdot \quad \cdot \quad \cdot$$

$$\cdot \quad \cdot \quad \cdot$$

$$n^3 x - 1 < [n^3 x] \leq n^3 x$$

Adding all these inequalities

इन सभी असमिकाओं को जोड़ने पर

$$(1^3 + 2^3 + 3^3 + \dots + n^3) x - n < [1^3 x] + [2^3 x] + \dots + [n^3 x] \leq (1^3 + 2^3 + \dots + n^3) x$$

$$\frac{n^2(n+1)^2}{4} x - n < \frac{[1^3 x] + [2^3 x] + \dots + [n^3 x]}{n^4} \leq \frac{\left(\frac{n(n+1)}{2}\right)^2 x}{n^4}$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^2 \frac{x}{4} - \frac{1}{n^3} < \lim_{n \rightarrow \infty} \frac{[1^3 x] + [2^3 x] + \dots + [n^3 x]}{n^4} \leq \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^2 \frac{x}{4}$$

$$\frac{x}{4} < \lim_{x \rightarrow \infty} \frac{[1^3 x] + [2^3 x] + \dots + [n^3 x]}{n^4} \leq \frac{x}{4}$$

$$\therefore \lim_{x \rightarrow \infty} \frac{[1^3 x] + [2^3 x] + \dots + [n^3 x]}{n^4} = \frac{x}{4}$$

25. $\lim_{x \rightarrow \infty} \frac{\log x^n - [x]}{[x]}$, $n \in \mathbb{N}$, (where $[x]$ denotes greatest integer less than or equal to x)

- (A) Has value -1 (B) Has value 0 (C) Has value 1 (D) Does not exist

$\lim_{x \rightarrow \infty} \frac{\log x^n - [x]}{[x]}$, $n \in \mathbb{N}$ (जहाँ $[x]$ महत्तम पूर्णांक x से कम या बराबर को प्रदर्शित करता है)

- (A) का मान -1 है (B) का मान 0 है (C) का मान 1 है (D) विद्यमान नहीं है

Ans. A

Sol. $\lim_{x \rightarrow \infty} \frac{\log x^n - [x]}{[x]} = \lim_{x \rightarrow \infty} \frac{\log x^n}{[x]} - \lim_{x \rightarrow \infty} \frac{[x]}{[x]} = 0 - 1 = -1$

26. Evaluate $\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+2n}} \right)$

$\lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+2n}} \right)$ का मान है -

(A) 1 (B) 1/2 (C) 0 (D) 2

Ans. D

Sol. Let $f(x) = \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2}} + \dots + \frac{1}{\sqrt{n^2}} \right) = \frac{2n+1}{\sqrt{n^2}}$

... (2n+1) terms

$$\lim_{n \rightarrow \infty} f(x) = 2$$

$g(x) = \left(\frac{1}{\sqrt{n^2+2n}} + \frac{1}{\sqrt{n^2+2n}} + \dots + \frac{1}{\sqrt{n^2+2n}} \right)$

...(2n+1) terms

$$\lim_{n \rightarrow \infty} f(x) = 2$$

$$\therefore \lim_{n \rightarrow \infty} \left(\frac{1}{\sqrt{n^2}} + \frac{1}{\sqrt{n^2+1}} + \frac{1}{\sqrt{n^2+2}} + \dots + \frac{1}{\sqrt{n^2+2n}} \right) = 2$$

27. $\lim_{x \rightarrow \infty} \frac{3}{x} \left[\frac{x}{4} \right] = \frac{p}{q}$ (where $[.]$ denotes greatest integer function), then $p + q$ (where p, q are relative prime) is :

$\lim_{x \rightarrow \infty} \frac{3}{x} \left[\frac{x}{4} \right] = \frac{p}{q}$ है (जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है) तो $p + q$ (जहाँ p, q परस्पर अभाज्य संख्यायें हैं) का मान होगा :

(A) 2 (B) 7 (C) 5 (D) 6

Ans. B

Sol. $\lim_{x \rightarrow \infty} \frac{3}{x} \left[\frac{x}{4} \right] = \frac{p}{q}$

$$\therefore \frac{x}{4} - 1 \leq \left[\frac{x}{4} \right] < \frac{x}{4}$$

$$\Rightarrow \frac{3}{x} \frac{(x-4)}{4} \leq \left[\frac{x}{4} \right] \cdot \frac{3}{x} < \frac{3}{4}$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{3}{4} \left(1 - \frac{4}{x} \right) \leq \lim_{x \rightarrow \infty} \left[\frac{x}{4} \right] \cdot \frac{3}{x} < \frac{3}{4}$$

$$\Rightarrow \frac{p}{q} = \frac{3}{4}$$

$$\Rightarrow p + q = 7$$

28. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a positive increasing function with $\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$. Then $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} =$

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ एक धन वर्धमान फलन है जिसमें $\lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$ है। तो $\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} =$

- (A) 1 (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) 3

Ans. A

Sol. $x < 2x < 3x$

$$f(x) < f(2x) < f(3x)$$

$$\frac{f(x)}{f(x)} < \frac{f(2x)}{f(x)} < \frac{f(3x)}{f(x)}$$

$$\lim_{x \rightarrow \infty} \frac{f(x)}{f(x)} = 1 \text{ and } \lim_{x \rightarrow \infty} \frac{f(3x)}{f(x)} = 1$$

Hence by Sandwich theorem

$$\lim_{x \rightarrow \infty} \frac{f(2x)}{f(x)} = 1$$

29. For each $t \in \mathbb{R}$, let $[t]$ be the greatest integer less than or equal to t . Then $\lim_{x \rightarrow 0^+} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{15}{x} \right] \right)$

- (A) is equal to 120 (B) does not exist (in $\mathbb{R}$)
(C) is equal to 0 (D) is equal to 15

प्रत्येक $t \in \mathbb{R}$ के लिए माना $[t]$, t से छोटा महतम पूर्णांक है, तो $\lim_{x \rightarrow 0^+} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{15}{x} \right] \right)$

- (A) 120 के बराबर है (B) ($\mathbb{R}$ में) इसका अस्तित्व नहीं है
(C) 0 के बराबर है (D) 15 के बराबर है

Ans. A

Sol. $x - 1 < [x] \leq x$

$$\left(\frac{1}{x}-1\right) < \left[\frac{1}{x}\right] \leq \frac{1}{x}$$

$$x\left(\frac{1}{x}-1\right) < x\left[\frac{1}{x}\right] \leq \frac{x}{x}$$

use sandwich

$$\lim_{x \rightarrow 0} x \left[\frac{1}{x}\right] = 1$$

$$\text{Similarly } \lim_{x \rightarrow 0} x \left[\frac{2}{x}\right] = 2$$

$\vdots$
 $\vdots$
 $\vdots$

$$\lim_{x \rightarrow 0} x \left[\frac{15}{x}\right] = 15$$

$$\text{Ans} = 1 + 2 + \dots + 15 = 120$$

QUESTIONS BASED ON DERIVATIVES

30. If $f(1) = 1$, $f'(1) = 2$, then $\lim_{x \rightarrow 1} \frac{\sqrt{f(x)} - 1}{\sqrt{x} - 1} =$

यदि $f(1) = 1$, $f'(1) = 2$, तो $\lim_{x \rightarrow 1} \frac{\sqrt{f(x)} - 1}{\sqrt{x} - 1} =$

(A) 2

(B) 1

(C) 3

(D) 4

Ans. A

Sol. $y = \lim_{x \rightarrow 1} \frac{\sqrt{l(x)} - 1}{\sqrt{x} - 1}$

$$\Rightarrow y = \lim_{x \rightarrow 1} \frac{(\sqrt{l(x)} - 1)(\sqrt{l(x)} + 1)}{(\sqrt{x} - 1)(\sqrt{x} + 1)} \cdot \frac{(\sqrt{x} + 1)}{(\sqrt{l(x)} + 1)}$$

$$\Rightarrow y = \lim_{x \rightarrow 1} \frac{l(x) - 1}{x - 1} \cdot \frac{\sqrt{x} + 1}{\sqrt{l(x)} + 1}$$

$$\Rightarrow y = \lim_{x \rightarrow 1} \frac{l(x) - l(1)}{x - 1} \cdot \lim_{x \rightarrow 1} \frac{\sqrt{x} + 1}{\sqrt{l(x)} + 1}$$

$$\Rightarrow y = l'(1) \cdot \frac{2}{\sqrt{l(1)+1}} \Rightarrow y = 2 \cdot \frac{2}{2} = 2$$

Aliter : Applying L-Hospital's rule.

31. If function $f(x)$ is differentiable at $x = a$ then $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$:

यदि फलन $f(x)$, $x = a$ पर अवकलनीय है, तो $\lim_{x \rightarrow a} \frac{x^2 f(a) - a^2 f(x)}{x - a}$ बराबर है :

(A) $2a f(a) + a^2 f'(a)$ (B) $-a^2 f'(a)$ (C) $af(a) - a^2 f'(a)$ (D) $2af(a) - a^2 f'(a)$

Ans. D

Sol. $\lim_{x \rightarrow a} \frac{x^2 l(a) - a^2 l(x)}{x - a}$ $\left[\frac{0}{0} \text{ form} \right]$

Use L'Hospital rule

$$= \lim_{x \rightarrow a} \frac{2x l(a) - a^2 l'(x)}{1}$$

$$= 2af(a) - a^2 f'(a)$$

LIMIT BASED ON EXPANSION

32. $\lim_{x \rightarrow 0} \frac{(\cos x) \ln(1+x) - x \cos x - \ln(1+x) + x}{x^4}$ is equal to :

$\lim_{x \rightarrow 0} \frac{(\cos x) \ln(1+x) - x \cos x - \ln(1+x) + x}{x^4}$ का मान होगा :

(A) 0 (B) $\frac{1}{2}$ (C) $\frac{1}{4}$ (D) $\frac{1}{8}$

Ans. C

Sol. $\lim_{x \rightarrow 0} \frac{\cos x \cdot \ln(1+x) - x \cos x - \ln(1+x) + x}{x^4} = \lim_{x \rightarrow 0} \frac{(\ln(1+x) - x)(\cos x - 1)}{x^4} = \left(\frac{-1}{2} \right) \left(\frac{-1}{2} \right) = \frac{1}{4}$

33. $\lim_{x \rightarrow 0} \frac{e^{-x^2/2} - \cos x}{x^3 \sin x} =$

(A) $1/4$ (B) $1/6$ (C) $1/12$ (D) $1/8$

Ans. C

Sol.

$$\lim_{x \rightarrow 0} \frac{e^{\frac{-x^2}{2}} - \cos x}{x^3 \sin x} = \lim_{x \rightarrow 0} \frac{\left[1 - \frac{x^2}{2} + \frac{x^4}{4 \cdot 2!} - \dots\right] - \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right]}{x^3 \left[x - \frac{x^3}{3!} + \dots\right]}$$

$$= \lim_{x \rightarrow 0} \frac{x^4 \left[\frac{1}{8} - \frac{1}{4!}\right] - x^6 \left[\frac{1}{8 \cdot 3!} - \frac{1}{6!}\right] + \dots}{x^4 \left[1 - \frac{x^2}{3!} + \dots\right]} = \left[\frac{1}{8} - \frac{1}{24}\right] = \frac{1}{12}$$

34. If $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$, then the values of a and b, are :

यदि $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$, तो a व b का मान होगा :

(A) $a \in \mathbb{R}, b \in \mathbb{R}$

(B) $a = 1, b \in \mathbb{R}$

(C) $a \in \mathbb{R}, b = 2$

(D) $a = 1$ and $b = 2$

Ans. B

Sol. Since, $\lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} + \frac{b}{x^2}\right)^{2x} = e^2$

$$\therefore \lim_{x \rightarrow \infty} \left[\left(1 + \frac{ax+b}{x^2}\right)^{\frac{x^2}{ax+b}} \right]^{\frac{2(ax+b)}{x}} = e^2$$

$$\Rightarrow \lim_{x \rightarrow \infty} e^{\frac{2(ax+b)}{x}} = e^2$$

$$\Rightarrow \lim_{x \rightarrow \infty} \frac{2(ax+b)}{x} = 2 \Rightarrow 2a = 2 \Rightarrow a = 1$$

Thus $a = 1$ and $b \in \mathbb{R}$

35. The value of Limit $\frac{\cos(\sin x) - \cos x}{x^4}$ is equal to :

Limit $\frac{\cos(\sin x) - \cos x}{x^4}$ का मान है :

(A) 1/5

(B) 1/6

(C) 1/4

(D) 1/2

Ans. B

Sol. $\lim_{x \rightarrow 0} \frac{\cos(\sin x) - \cos x}{x^4} = \lim_{x \rightarrow 0} \frac{\cos\left(x - \frac{x^3}{3!} + \dots\right) - \cos x}{x^4}$

$$= \lim_{x \rightarrow 0} \frac{\left[1 - \frac{\left(x - \frac{x^3}{6} + \dots\right)^2}{2!} + \frac{\left(x - \frac{x^3}{6} + \dots\right)^4}{4!} - \dots\right] - \left[1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots\right]}{x^4}$$

$$= \lim_{x \rightarrow 0} \frac{x^4 \left(\frac{1}{2} \cdot \frac{2}{6} + \frac{1}{24}\right) - \frac{x^4}{24} + \dots}{x^4} = \frac{1}{6} + \frac{1}{24} - \frac{1}{24} = \frac{1}{6}$$

36. The integer 'n' for which $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite non-zero number is :

पूर्णांक 'n' जिसके लिए $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ एक परिमित अशून्य संख्या है, है :

(A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. $\lim_{x \rightarrow 0} \frac{(\cos x - 1)(\cos x - e^x)}{x^n} = 1 \text{ (non zero finite)}$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\left(-\frac{x^2}{2} + \frac{x^4}{4} - \frac{x^6}{6} + \dots\right) \left(1 - \frac{x^2}{2} + \frac{x^4}{4} - \dots\right) - \left(1 + \frac{x^2}{2} + \frac{x^3}{3} + \dots\right)}{x^n} = 1$$

$$\Rightarrow n = 3$$

EVALUATION OF LIMITS USING STANDARD OPERATION

1. $\lim_{x \rightarrow 1} \frac{x^2 - x \ln x + \ln x - 1}{x - 1}$

Ans. 2

Sol.

2. $\lim_{x \rightarrow 1} \frac{\left[\sum_{K=1}^{100} x^K \right] - 100}{x - 1}$

Ans. 5050

Sol. $\lim_{x \rightarrow 1} \frac{\left[\sum_{K=1}^{100} x^K - 100 \right]}{x - 1}$

$$= \lim_{x \rightarrow 1} \left[\frac{x + x^2 + x^3 + \dots + x^{100} - 100}{x - 1} \right]$$

$$= \lim_{x \rightarrow 1} \left[\frac{1 + 2x + 3x^2 + \dots + 100x^{99}}{1} \right]$$

(using L' Hospital)

$$= 5050$$

3. $\lim_{x \rightarrow \frac{\pi}{4}} \frac{1 - \tan x}{1 - \sqrt{2} \sin x}$

Ans. 2

Sol. Put $x = \frac{\pi}{4} + h$

$$\therefore x \rightarrow \frac{\pi}{4} \Rightarrow h \rightarrow 0$$

$$\lim_{h \rightarrow 0} \frac{1 - \tan\left(\frac{\pi}{4} + h\right)}{1 - \sqrt{2} \sin\left(\frac{\pi}{4} + h\right)} = \lim_{h \rightarrow 0} \frac{1 - \frac{1 + \tan h}{1 - \tan h}}{1 - \sin h - \cos h} = \lim_{h \rightarrow 0} \frac{\frac{-2 \tan h}{1 - \tan h}}{2 \sin^2 \frac{h}{2} - 2 \sin \frac{h}{2} \cos \frac{h}{2}}$$

$$= \lim_{h \rightarrow 0} \frac{-2 \tan h}{2 \sin \frac{h}{2} \left[\sin \frac{h}{2} - \cos \frac{h}{2} \right]} \cdot \frac{1}{(1 - \tanh)} = \lim_{h \rightarrow 0} \frac{-2 \frac{\tanh}{h}}{\frac{\sin \frac{h}{2}}{\frac{h}{2}} \left[\sin \frac{h}{2} - \cos \frac{h}{2} \right]} \cdot \frac{1}{(1 - \tanh)} = \frac{-2}{-1} = 2.$$

4. Find the sum of an infinite geometric series whose first term is the limit of the function $f(x) = \frac{\tan x - \sin x}{\sin^3 x}$ as $x \rightarrow 0$ and whose common ratio is the limit of the function $g(x) = \frac{1 - \sqrt{x}}{(\cos^{-1} x)^2}$ as $x \rightarrow 1$.

ऐसे अनन्त पदों वाले गुणोत्तर श्रेणी का योग ज्ञात करें जिसका प्रथम पद फलन $f(x) = \frac{\tan x - \sin x}{\sin^3 x}$ की $x \rightarrow 0$ पर सीमा है तथा

सर्वानुपात फलन $g(x) = \frac{1 - \sqrt{x}}{(\cos^{-1} x)^2}$ की $x \rightarrow 1$ पर सीमा है।

Ans. $a = \frac{1}{2}; r = \frac{1}{4}; S = \frac{2}{3}$

Sol. $a = \lim_{x \rightarrow 0} \frac{\tan x - \sin x}{\sin^3 x} = \lim_{x \rightarrow 0} \frac{1 - \cos x}{\sin^2 x \cdot \cos x}$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{4 \sin^2 \frac{x}{2} \cos^2 \frac{x}{2} \cos x} = \frac{1}{2}$$

$$r = \lim_{x \rightarrow 1} \frac{1 - \sqrt{x}}{(\cos^{-1} x)^2} \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{x \rightarrow 1} \frac{1 - x}{(1 + \sqrt{x})(\cos^{-1} x)^2} = \frac{1}{2} \lim_{\theta \rightarrow 0} \frac{2 \sin^2 \frac{\theta}{2}}{\theta^2}$$

$$= \frac{1}{4}$$

$$\Rightarrow S_{\infty} = \frac{a}{1 - r} = \frac{\frac{1}{2}}{1 - \frac{1}{4}} = \frac{1 \cdot 4}{2 \cdot 3} = \frac{2}{3}$$

5. $\lim_{x \rightarrow \infty} (x - \ln \cosh x)$ where $\cosh t = \frac{e^t + e^{-t}}{2}$.

$$\lim_{x \rightarrow \infty} (x - \ln \cosh x) \text{ जहाँ पर } \cosh t = \frac{e^t + e^{-t}}{2}.$$

Ans. $\ln 2$

Sol. $\lim_{x \rightarrow \infty} (x - \ln \cosh x) \quad (\infty - \infty \text{ form})$

$$\begin{aligned}
&= \lim_{x \rightarrow \infty} \left(x - \ln \left(\frac{e^x + e^{-x}}{2} \right) \right) \\
&= \lim_{x \rightarrow \infty} \left(x - \ln \left(\frac{e^{2x} + 1}{2e^x} \right) \right) \\
&= \lim_{x \rightarrow \infty} \left(x - \ln(e^{2x} + 1) + \ln 2 + \ln(e^x) \right) \\
&= \lim_{x \rightarrow \infty} \left(2x - \ln(e^{2x} + 1) + \ln 2 \right) \\
&= \lim_{x \rightarrow \infty} \left[\ln \left(\frac{e^{2x}}{e^{2x} + 1} \right) + \ln 2 \right] \\
&= \ln 2
\end{aligned}$$

6. $\lim_{x \rightarrow \frac{3\pi}{4}} \frac{1 + \sqrt[3]{\tan x}}{1 - 2 \cos^2 x}$

Ans. $-\frac{1}{3}$

Sol. $\lim_{x \rightarrow \frac{3\pi}{4}} \frac{1 + (\tan x)^{\frac{1}{3}}}{1 - 2 \cos^2 x} = \lim_{x \rightarrow \frac{3\pi}{4}} \frac{1 + (\tan x)^{\frac{1}{3}}}{-\cos 2x}$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{1 + \left[\tan \left(\frac{3\pi}{4} + h \right) \right]^{\frac{1}{3}}}{-\cos \left(\frac{3\pi}{2} + 2h \right)} = \lim_{h \rightarrow 0} \frac{1 - \left(\tan \left(\frac{\pi}{4} - h \right) \right)^{\frac{1}{3}}}{-2h} \\
&= \lim_{h \rightarrow 0} \frac{1^{\frac{1}{3}} - \left(\tan \left(\frac{\pi}{4} - h \right) \right)^{\frac{1}{3}}}{-2h} = -\frac{1}{2} \lim_{h \rightarrow 0} \frac{1 - \tan \left(\frac{\pi}{4} - h \right)}{3 \cdot h} = -\frac{1}{3}
\end{aligned}$$

$$\left\{ \because a^3 + b^3 = (a + b)(a^2 - ab + b^2) \right\}$$

7. $\lim_{x \rightarrow 0} \frac{8}{x^8} \left[1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right]$

Ans. $\frac{1}{32}$

Sol.
$$\lim_{x \rightarrow 0} \frac{8}{x^8} \left[1 - \cos \frac{x^2}{2} - \cos \frac{x^2}{4} + \cos \frac{x^2}{2} \cos \frac{x^2}{4} \right]$$

$$= \lim_{x \rightarrow 0} \frac{8}{x^8} \left(1 - \cos \frac{x^2}{2} \right) \left(-\cos \frac{x^2}{4} + 1 \right)$$

$$= \lim_{x \rightarrow 0} \frac{8}{x^8} 2 \sin^2 \frac{x^2}{4} 2 \sin^2 \frac{x^2}{8}$$

$$= 32 \lim_{x \rightarrow 0} \left(\frac{\sin \frac{x^2}{4}}{\frac{x^2}{4}} \right)^2 \left(\frac{\sin \frac{x^2}{8}}{\frac{x^2}{8}} \right)^2 \cdot \frac{1}{16 \cdot 64}$$

$$= \frac{1}{32}$$

8. If $l = \lim_{n \rightarrow \infty} \sum_{r=2}^n \left((r+1) \sin \frac{\pi}{r+1} - r \sin \frac{\pi}{r} \right)$ then find $\{l\}$. (where $\{ \}$ denotes the fractional part function)

यदि $l = \lim_{n \rightarrow \infty} \sum_{r=2}^n \left((r+1) \sin \frac{\pi}{r+1} - r \sin \frac{\pi}{r} \right)$ तो $\{l\}$ मान ज्ञात करो। (जहाँ पर $\{ \}$ भिन्नांश फलन है)

Ans. $\pi - 3$

Sol.
$$\ell = \lim_{n \rightarrow \infty} \sum_{r=2}^n \left((r+1) \sin \frac{\pi}{r+1} - r \sin \frac{\pi}{r} \right)$$

$$\ell = \lim_{n \rightarrow \infty} \sum_{r=2}^n \left[r \left(\sin \frac{\pi}{r+1} - \sin \frac{\pi}{r} \right) + \sin \left(\frac{\pi}{r+1} \right) \right]$$

$$= \lim_{n \rightarrow \infty} \left(3 \sin \frac{\pi}{3} - 2 \sin \frac{\pi}{2} + 4 \sin \frac{\pi}{4} - 3 \sin \frac{\pi}{3} + \dots + n \sin \frac{\pi}{n} - (n-1) \sin \frac{\pi}{n-1} \right. \\ \left. + (n+1) \sin \frac{\pi}{n+1} - n \sin \frac{\pi}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \left((n+1) \sin \frac{\pi}{n+1} - 2 \right)$$

$$= \lim_{n \rightarrow \infty} \left[\pi \frac{\sin \left(\frac{\pi}{n+1} \right)}{\frac{\pi}{n+1}} - 2 \right] = \pi - 2$$

$$\Rightarrow \{\ell\} = \pi - 3$$

9.
$$\lim_{x \rightarrow 1} \frac{(\ln(1+x) - \ln 2)(3 \cdot 4^{x-1} - 3x)}{[(7+x)^{\frac{1}{3}} - (1+3x)^{\frac{1}{2}}] \cdot \sin(x-1)}$$

Ans.
$$-\frac{9}{4} \ln \frac{4}{e}$$

Sol.
$$\lim_{x \rightarrow 1} \frac{\ln\left(\frac{1+x}{2}\right) \times 3 \cdot (4^{x-1} - x)}{[(7+x)^{\frac{1}{3}} - (1+3x)^{\frac{1}{2}}] \sin(x-1)}$$

$$= \lim_{h \rightarrow 0} \frac{\ln\left(\frac{2+h}{2}\right) \times 3(4^h - (1+h))}{[(8+h)^{\frac{1}{3}} - (4+3h)^{1/2}] \cdot \sin h}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{\ln\left(1+\frac{h}{2}\right)}{2(h/2)} \cdot 3 \cdot (4^h - (1+h))}{2\left\{\left[1+\frac{h}{8} \times \frac{1}{3} + \dots\right] - \left[1+\frac{3h}{4} \times \frac{1}{2} + \dots\right]\right\} \frac{\sin h}{h}}$$

$$= \lim_{h \rightarrow 0} \frac{\frac{3}{4} \left[\frac{4^h - 1}{h} - 1 \right]}{\frac{1}{h} \left[\frac{h}{24} - \frac{3h}{8} \right]} = \frac{-9}{4} \ln \frac{4}{e}$$

10.
$$\lim_{x \rightarrow 0} \frac{27^x - 9^x - 3^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}$$

Ans.
$$8\sqrt{2}(\ln 3)^2$$

Sol.
$$\lim_{x \rightarrow 0} \frac{3^{3x} - 3^{2x} - 3^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}} = \lim_{x \rightarrow 0} \frac{(3^x - 1)^2 (3^x + 1)}{\sqrt{2} \left(1 - \cos \frac{x}{2}\right)}$$

$$= \sqrt{2} \lim_{x \rightarrow 0} \frac{(3^x - 1)^2 \frac{x^2}{16}}{2 \sin^2 \frac{x}{4} \frac{x^2}{16}}$$

$$= 8\sqrt{2} \lim_{x \rightarrow 0} \frac{(3^x - 1)^2}{x^2} = 8\sqrt{2} \lim_{x \rightarrow 0} \frac{(e^{x \ln 3} - 1)^2}{x^2}$$

$$= 8\sqrt{2} (\ln 3)^2$$

11. Let $P_n = a^{P_{n-1}} - 1$, $\forall n = 2, 3, \dots$ and Let $P_1 = a^x - 1$ where $a \in \mathbb{R}^+$ then evaluate $\lim_{x \rightarrow 0} \frac{P_n}{x}$.

माना कि $P_n = a^{P_{n-1}} - 1$, $\forall n = 2, 3, \dots$ और $P_1 = a^x - 1$ जहाँ पर $a \in \mathbb{R}^+$ तो $\lim_{x \rightarrow 0} \frac{P_n}{x}$ का मान ज्ञात करें।

Ans. $(\ln a)^n$

Sol. $P_n = a^{P_{n-1}} - 1 \quad \forall n = 2, 3, \dots$

$$P_1 = a^x - 1; a \in \mathbb{R}^+ = e^{x \ln a} - 1$$

$$P_2 = a^{a^x - 1} - 1 = e^{(a^x - 1) \ln a} - 1$$

$$P_3 = a^{a^{a^x - 1} - 1} - 1 = a^{P_2} - 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{P_n}{x} = \lim_{x \rightarrow 0} \frac{a^{P_n} - 1}{P_n} \cdot \frac{P_{n-1}}{x}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{a^{P_{n-1}} - 1}{P_{n-1}} \cdot \frac{a^{P_{n-2}} - 1}{P_{n-2}} \cdot \frac{a^{P_{n-3}} - 1}{P_{n-3}} \cdot \dots \cdot \frac{a^{P_2} - 1}{P_2} \cdot \frac{a^{P_1} - 1}{P_1} \cdot \frac{a^x - 1}{x}$$

$$= (\ln(a))^n$$

12. $\lim_{y \rightarrow 0} \left[\lim_{x \rightarrow \infty} \frac{\exp\left(x \ln\left(1 + \frac{ay}{x}\right)\right) - \exp\left(x \ln\left(1 + \frac{by}{x}\right)\right)}{y} \right]$

Ans. $a - b$

Sol. $\lim_{y \rightarrow 0} \left(\lim_{x \rightarrow \infty} \frac{\exp\left(x \ln\left(1 + \frac{ay}{x}\right)\right) - \exp\left(x \ln\left(1 + \frac{by}{x}\right)\right)}{y} \right) = \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow \infty} \frac{\left(1 + \frac{ay}{x}\right)^x - \left(1 + \frac{by}{x}\right)^x}{y} \right)$

by expansion प्रसार करने पर

$$= \lim_{y \rightarrow 0} \left(\lim_{x \rightarrow \infty} \frac{\left(1 + ay + \frac{x(x-1)}{2!} \cdot \frac{a^2 y^2}{x^2} + \dots\right) - \left(1 + by + \frac{x(x-1)}{2!} \cdot \frac{b^2 y^2}{x^2} + \dots\right)}{y} \right)$$

$$= \lim_{y \rightarrow 0} \left[\frac{y(a-b) + \frac{y^2}{2}(a^2 - b^2) + \dots}{y} \right] = a - b$$

13. Let $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ be sequences such that

(i) $a_n + b_n + c_n = 2n + 1$; (ii) $a_n b_n + b_n c_n + c_n a_n = 2n - 1$; (iii) $a_n b_n c_n = -1$; (iv) $a_n < b_n < c_n$

Then find the value of $\lim_{n \rightarrow \infty} (na_n)$.

यदि $\{a_n\}$, $\{b_n\}$, $\{c_n\}$ इस प्रकार की श्रेणी है कि

(i) $a_n + b_n + c_n = 2n + 1$; (ii) $a_n b_n + b_n c_n + c_n a_n = 2n - 1$; (iii) $a_n b_n c_n = -1$; (iv) $a_n < b_n < c_n$

तो $\lim_{n \rightarrow \infty} (na_n)$ का मान ज्ञात करें।

Ans. $-1/2$

Sol. $x^3 - (2n+1)x^2 + (2n-1)x + 1 = 0$

$$x = 1 \quad \text{or} \quad x = n - \sqrt{n^2 + 1} \quad \text{or} \quad x = n + \sqrt{n^2 + 1}$$

$$\therefore a_n = n - \sqrt{n^2 + 1}$$

$$\Rightarrow a_n = \frac{-1}{n + \sqrt{n^2 + 1}}$$

$$\Rightarrow \lim_{n \rightarrow \infty} na_n = \lim_{n \rightarrow \infty} \frac{-1}{1 + \sqrt{1 + \frac{1}{n^2}}} = -\frac{1}{2}$$

14. $\lim_{x \rightarrow 0} \left(\frac{x-1+\cos x}{x} \right)^{\frac{1}{x}}$

Ans. $e^{-1/2}$

Sol. $\lim_{x \rightarrow 0} \left(\frac{x-1+\cos x}{x} \right)^{\frac{1}{x}} \quad (1^\infty \text{ form})$

$$= \exp \left[\lim_{x \rightarrow 0} \left(\frac{-1+\cos x}{x^2} \right) \right] = \exp \left[\lim_{x \rightarrow 0} \frac{-2 \sin^2 \frac{x}{2}}{\frac{x^2}{4} \cdot 4} \right] = e^{-\frac{1}{2}}$$

15. If $n \in \mathbb{N}$ and $a_n = 2^2 + 4^2 + 6^2 + \dots + (2n)^2$ and $b_n = 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2$. Find the value

$$\lim_{n \rightarrow \infty} \frac{\sqrt{a_n} - \sqrt{b_n}}{\sqrt{n}}.$$

यदि $n \in \mathbb{N}$ तथा $a_n = 2^2 + 4^2 + 6^2 + \dots + (2n)^2$ और $b_n = 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2$ तो $\lim_{n \rightarrow \infty} \frac{\sqrt{a_n} - \sqrt{b_n}}{\sqrt{n}}$ का

मान ज्ञात करो।

Ans. $\sqrt{3}/2$

Sol. $a_n = 2^2 + 4^2 + 6^2 + \dots + (2n)^2 = 2^2(1^2 + 2^2 + \dots + n^2)$

$$= 4 \frac{n(n+1)(2n+1)}{6} = \frac{2}{3}n(n+1)(2n+1)$$

$$b_n = 1^2 + 3^2 + 5^2 + \dots + (2n-1)^2$$

$$= \frac{2n(2n+1)(4n+1)}{6} - \frac{4n(n+1)(2n+1)}{6}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{\sqrt{a_n} - \sqrt{b_n}}{\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{(a_n - b_n)}{\sqrt{n}(\sqrt{a_n} + \sqrt{b_n})}$$

$$= \lim_{n \rightarrow \infty} -2 \frac{\frac{4n(2n+1)(4n+1)}{6} + \frac{8}{6}n(n+1)(2n+1)}{\sqrt{n} \left(\sqrt{\frac{4}{6}n(n+1)(2n+1)} + \sqrt{\frac{2n(2n+1)(4n+1)}{6} - \frac{4}{6}n(n+1)(2n+1)} \right)}$$

$$= -\lim_{n \rightarrow \infty} \frac{n^3 \left[\frac{5}{6} \left(2 + \frac{1}{n} \right) \left(4 + \frac{1}{n} \right) - \frac{8}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) \right]}{n^2 \left(\sqrt{\frac{4}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right)} + \sqrt{\frac{2 \left(2 + \frac{1}{n} \right) \left(4 + \frac{1}{n} \right)}{6}} - \frac{4}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) \right)}$$

$$= -\lim_{n \rightarrow \infty} \frac{n \left[\frac{2}{6} \left(2 + \frac{1}{n} \right) \left(4 + \frac{1}{n} \right) - \frac{8}{6} \left(1 + \frac{1}{n} \right) \left(2 + \frac{1}{n} \right) \right]}{\left(\sqrt{\frac{8}{6}} + \sqrt{\frac{16}{6} - \frac{8}{6}} \right)}$$

$$= -\frac{1}{2} \sqrt{\frac{6}{8}} \lim_{y \rightarrow 0} \frac{\frac{1}{3}(2+y)(4+y) - \frac{4}{3}(1+y)(2+y)}{y}$$

$$= \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

16. If $|x| < 1$, then $\lim_{n \rightarrow \infty} \left((1+x)(1+x^2) \dots (1+x^{2^n}) \right)$ equal to :

यदि $|x| < 1$ है, तो $\lim_{n \rightarrow \infty} \left((1+x)(1+x^2) \dots (1+x^{2^n}) \right)$ मान बराबर होगा :

Ans. $\frac{1}{1-x}$

Sol. Given $|x| < 1$

$$\lim_{n \rightarrow \infty} \left((1+x)(1+x^2) \dots (1+x^{2^n}) \right)$$

$$= \lim_{n \rightarrow \infty} 1 + x + x^2 + \dots + x^{2^{n+1}}$$

$$= \frac{1}{1-x}$$

BASED ON SPECIAL SIGNS LIKE GIF, MOD, SGN ETC.

17. Let $f(x) = \begin{cases} -5 & , \quad -\infty < x \leq -1 \\ x-4 & , \quad -1 < x \leq 6 \\ 2 & , \quad 6 < x < \infty \end{cases}$ then $\lim_{x \rightarrow n} \frac{[x]^2 - 13[x] + 42}{(x-7)(x-6)}$ is equal to (where 'n' is the number of non zero

integers in the range of $f(x)$, $[.]$ denotes greatest integer function)

माना $f(x) = \begin{cases} -5 & , \quad -\infty < x \leq -1 \\ x-4 & , \quad -1 < x \leq 6 \\ 2 & , \quad 6 < x < \infty \end{cases}$ तो $\lim_{x \rightarrow n} \frac{[x]^2 - 13[x] + 42}{(x-7)(x-6)}$ का मान होगा (जहाँ n, $f(x)$ के परिसर में अशून्य पूर्णाको

की संख्या है तथा $[.]$ महत्तम पूर्णांक फलन दर्शाता है)

Ans. 0

Sol. $\lim_{x \rightarrow \pi/4} \frac{\sqrt{1 - \sqrt{\sin 2x}}}{\pi - 4x}$

$$\text{LHL} = \lim_{h \rightarrow 0} \frac{\sqrt{1 - \sqrt{\sin\left(\frac{\pi}{2} - 2h\right)}}}{\pi - \pi + 4h}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{\sqrt{1 - \sqrt{\cos 2h}}}{4h} \times \frac{\sqrt{1 + \sqrt{\cos 2h}}}{\sqrt{1 + \sqrt{\cos 2h}}} \\
&= \lim_{h \rightarrow 0} \frac{\sqrt{1 - \cos 2h}}{4h} \times \frac{1}{\sqrt{1 + \sqrt{\cos 2h}}} \\
&= \frac{\sqrt{2} |\sin h|}{4h \times \sqrt{2}} = \frac{1}{4}
\end{aligned}$$

Similarly RHL = $\frac{-1}{4}$

Hence LHL $\neq$ RHL $\therefore$ limit does not exist.

18. The value of $\lim_{x \rightarrow 0} 10 \left[\frac{\sin^{-1} x}{x} \right] + 5 \left[\frac{\sin^{-1} 2x}{x} \right] + 20 \left[\frac{x}{\tan^{-1} x} \right] + 10 \left[\frac{x}{\tan^{-1} 2x} \right]$ (where $[.]$ denotes greatest integer function) is equal to –

$\lim_{x \rightarrow 0} 10 \left[\frac{\sin^{-1} x}{x} \right] + 5 \left[\frac{\sin^{-1} 2x}{x} \right] + 20 \left[\frac{x}{\tan^{-1} x} \right] + 10 \left[\frac{x}{\tan^{-1} 2x} \right]$ का मान होगा (जहाँ $[.]$ महत्तम पूर्णांक फलन दर्शाता है)

Ans. 40

Sol. $\lim_{x \rightarrow 0} 10 \left[\frac{\sin^{-1} x}{x} \right] + 5 \left[\frac{\sin^{-1} 2x}{x} \right] + 20 \left[\frac{x}{\tan^{-1} x} \right] + 10 \left[\frac{x}{\tan^{-1} 2x} \right] = 10 + 10 + 20 + 0 = 40$

$$\left\{ \because \frac{\sin^{-1} x}{x} > 1, \frac{x}{\tan^{-1} x} > 1 \right\}$$

19. (a) $\lim_{x \rightarrow \frac{1}{\sqrt{2}}} \frac{\cos^{-1}(2x\sqrt{1-x^2})}{x - \frac{1}{\sqrt{2}}}$; (b) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{1 - \sqrt{\sin 2x}}}{\pi - 4x}$; (c) $\lim_{x \rightarrow -7} \frac{[x]^2 + 15[x] + 56}{\sin(x+7)\sin(x+8)}$
where $[.]$ denotes the greatest integer function

(a) $\lim_{x \rightarrow \frac{1}{\sqrt{2}}} \frac{\cos^{-1}(2x\sqrt{1-x^2})}{x - \frac{1}{\sqrt{2}}}$; (b) $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sqrt{1 - \sqrt{\sin 2x}}}{\pi - 4x}$; (c) $\lim_{x \rightarrow -7} \frac{[x]^2 + 15[x] + 56}{\sin(x+7)\sin(x+8)}$
जहाँ $[.]$ महत्तम पूर्णांक फलन है

Ans. (a) DNE (b) DNE (c) 0

Sol. $\lim_{x \rightarrow \frac{1}{2}} \frac{\cos^{-1}(2x\sqrt{1-x^2})}{x - \frac{1}{\sqrt{2}}} = \lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\cos^{-1}(\sin 2\theta)}{\sin \theta - \frac{1}{\sqrt{2}}}$ if $x = \sin \theta$

$$\text{RHL} = \lim_{h \rightarrow 0} \frac{\cos^{-1}\left(\sin\left(\frac{\pi}{2} + 2h\right)\right)}{\sin\left(\frac{\pi}{4} + h\right) - \frac{1}{\sqrt{2}}} = \lim_{h \rightarrow 0} \frac{2h\sqrt{2}}{\cosh + \sinh - 1}$$

$$= \lim_{h \rightarrow 0} \frac{2\sqrt{2}h}{h \cdot \left(\frac{\sinh}{h}\right) - \left(\frac{1 - \cosh}{h^2}\right) \cdot h^2} = \lim_{h \rightarrow 0} \frac{2\sqrt{2}}{\frac{\sinh}{h} - \left(\frac{1 - \cosh}{h^2}\right) \cdot h} = 2\sqrt{2}$$

$$\text{LHL} = \lim_{h \rightarrow 0} \frac{\cos^{-1}\left(\sin\left(\frac{\pi}{2} - 2h\right)\right)}{\sin\left(\frac{\pi}{4} - h\right) - \frac{1}{\sqrt{2}}} = \lim_{h \rightarrow 0} \frac{2h\sqrt{2}}{\cosh - \sinh - 1} = -2\sqrt{2}$$

$\Rightarrow$ limit DNE

20. $\lim_{x \rightarrow -\infty} \frac{(3x^4 + 2x^2)\sin \frac{1}{x} + |x|^3 + 5}{|x|^3 + |x|^2 + |x| + 1}$

Ans. -2

Sol. $\lim_{x \rightarrow \infty} \frac{(3x^4 + 2x^2)\sin \frac{1}{x} + |x|^3 + 5}{|x|^3 + |x|^2 + |x| + 1}$

$$= \lim_{x \rightarrow \infty} \frac{x^4 \left[3 + \frac{2}{x^2} \right] \frac{\sin \frac{1}{x}}{\frac{1}{x}} \cdot \frac{1}{x} - x^3 + 5}{x^3 \left(-1 - \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3} \right)}$$

$$= \lim_{x \rightarrow \infty} \frac{\left[3 + \frac{2}{x^2}\right] \frac{\sin \frac{1}{x}}{\frac{1}{x}} - 1 + \frac{5}{x^3}}{-1 - \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3}} = -2$$

21. Let $f(x) = \frac{x}{\sin x}, x > 0$ and $g(x) = x + 3, x < 1$
 $= 2 - x, x \leq 0$ $= x^2 - 2x - 2, 1 \leq x < 2$
 $= x - 5, x \geq 2$

find LHL and RHL of $g(f(x))$ at $x = 0$ and hence find $\lim_{x \rightarrow 0} g(f(x))$.

माना $f(x) = \frac{x}{\sin x}, x > 0$ और $g(x) = x + 3, x < 1$
 $= 2 - x, x \leq 0$ $= x^2 - 2x - 2, 1 \leq x < 2$
 $= x - 5, x \geq 2$

तो $g(f(x))$ का $x = 0$ पर वामपक्ष सीमा (LHL) तथा दाहिना पक्ष सीमा (RHL) ज्ञात करो। फलस्वरूप $\lim_{x \rightarrow 0} g(f(x))$ का मान ज्ञात करो।

Ans. $-3, -3, -3$

Sol. $f(x) = \begin{cases} \frac{x}{\sin x} & x > 0 \\ 2 - x & x \leq 0 \end{cases}$ $g(x) = \begin{cases} x + 3 & x < 1 \\ x^2 - 2x - 2 & 1 \leq x < 2 \\ x - 5 & x \geq 2 \end{cases}$

$$\text{LHL} = \lim_{h \rightarrow 0} f(0 - h) = \lim_{h \rightarrow 0} (2 + h) = 2$$

$$\text{RHL} = \lim_{h \rightarrow 0} f(0 + h) = \lim_{h \rightarrow 0} \frac{h}{\sinh} = 1$$

$$\text{LHL} = \lim_{h \rightarrow 0} g(f(-h)) = \lim_{h \rightarrow 0} g(2 + h) = \lim_{h \rightarrow 0} (2 + h - 5) = -3$$

$$\text{RHL} = \lim_{h \rightarrow 0} g(f(0 + h)) = \lim_{h \rightarrow 0} g\left(\frac{h}{\sinh}\right) = \lim_{h \rightarrow 0} \left(\frac{h}{\sinh}\right)^2 - 2\left(\frac{h}{\sinh}\right) - 2 = -3$$

$$\Rightarrow \lim_{x \rightarrow 0} g(f(x)) = -3$$

22. (a) $\lim_{x \rightarrow 0} \tan^{-1} \frac{a}{x^2}$, where $a \in \mathbb{R}$;

(b) Plot the graph of the function $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \tan^{-1} \frac{x}{t^2} \right)$

(a) $\lim_{x \rightarrow 0} \tan^{-1} \frac{a}{x^2}$, जहाँ पर $a \in \mathbb{R}$;

(b) फलन $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \tan^{-1} \frac{x}{t^2} \right)$ का रेखाचित्र खींचो ।

Ans. (a) $\pi/2$ if $a > 0$; 0 if $a = 0$ and $-\pi/2$ if $a < 0$; (b) $f(x) = |x|$

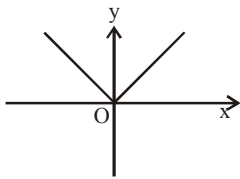
Sol. (a) $\lim_{x \rightarrow 0} \tan^{-1} \frac{a}{x^2}$; $a \in \mathbb{R}$

$$= \begin{cases} \frac{\pi}{2} & a > 0 \\ 0 & a = 0 \\ -\frac{\pi}{2} & a < 0 \end{cases}$$

(b) $f(x) = \lim_{t \rightarrow 0} \left(\frac{2x}{\pi} \tan^{-1} \frac{x}{t^2} \right)$

$$= \begin{cases} \frac{2x}{\pi} \cdot \frac{\pi}{2} & x > 0 \\ 0 & x = 0 \\ -\frac{2x}{\pi} \cdot \frac{\pi}{2} & x < 0 \end{cases}$$

$$= \begin{cases} x & x > 0 \\ 0 & x = 0 \\ -x & x < 0 \end{cases} = |x|$$



23. Find the value of $\lim_{n \rightarrow \infty} \left[\frac{2}{\pi} (n+1) \cos^{-1} \left(\frac{1}{n} \right) - n \right]$.

$\lim_{n \rightarrow \infty} \left[\frac{2}{\pi} (n+1) \cos^{-1} \left(\frac{1}{n} \right) - n \right]$ का मान ज्ञात करो ।

Ans. $1 - \frac{2}{\pi}$

Sol. $\lim_{n \rightarrow \infty} \left[\frac{2}{\pi} (n+1) \cos^{-1} \left(\frac{1}{n} \right) - n \right]$

$$= \lim_{n \rightarrow \infty} \left[(n+1) - \frac{2}{\pi} (n+1) \sin^{-1} \left(\frac{1}{n} \right) - n \right]$$

$$= \lim_{n \rightarrow \infty} \left[1 - \frac{2}{\pi} \left(1 + \frac{1}{n} \right) \frac{\sin^{-1} \frac{1}{n}}{\frac{1}{n}} \right]$$

$$= 1 - \frac{2}{\pi}$$

SANDWICH THEOREM

24. Using Sandwich theorem, evaluate

$$\lim_{n \rightarrow \infty} \frac{1}{1+n^2} + \frac{2}{2+n^2} + \dots + \frac{n}{n+n^2}$$

सैंडविच प्रमेय (Sandwich theorem) का प्रयोग करते हुये ज्ञात करो।

$$\lim_{n \rightarrow \infty} \frac{1}{1+n^2} + \frac{2}{2+n^2} + \dots + \frac{n}{n+n^2}$$

Ans. $1/2$

Sol. $\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r}{r+n^2}$

$$\because \sum_{r=1}^n \frac{r}{n^2+n} \leq \sum_{r=1}^n \frac{r}{r+n^2} \leq \sum_{r=1}^n \frac{r}{n^2+1}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \frac{n(n+1)}{2(n^2+n)} \leq \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r}{r+n^2} \leq \lim_{n \rightarrow \infty} \frac{n(n+1)}{2(n^2+1)}$$

$$\downarrow \qquad \qquad \qquad \downarrow$$

$$\frac{1}{2} \qquad \qquad \qquad \frac{1}{2}$$

$$\Rightarrow \lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{r}{r+n^2} = \frac{1}{2}$$

QUESTIONS BASED ON DERIVATIVES

25. $\lim_{h \rightarrow 0} \frac{\tan(a+2h) - 2\tan(a+h) + \tan a}{h^2} =$

Ans. $2 \sec^2 a \cdot \tan a$

Sol. $\lim_{h \rightarrow 0} \frac{\tan(a+2h) - 2\tan(a+h) + \tan a}{h^2}$
 $= \frac{d^2}{dx^2} (\tan x) \Big|_{x=a} = 2 \sec^2 a \tan a$

26. $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + 4h\right) - 4\sin\left(\frac{\pi}{3} + 3h\right) + 6\sin\left(\frac{\pi}{3} + 2h\right) - 4\sin\left(\frac{\pi}{3} + h\right) + \sin \frac{\pi}{3}}{h^4} =$

Ans. $\frac{\sqrt{3}}{2}$

Sol. $\lim_{h \rightarrow 0} \frac{\sin\left(\frac{\pi}{3} + 4h\right) - 4\sin\left(\frac{\pi}{3} + 3h\right) + 6\sin\left(\frac{\pi}{3} + 2h\right) - 4\sin\left(\frac{\pi}{3} + h\right) + \sin \frac{\pi}{3}}{h^4} = \frac{d^4}{dx^4} \sin(x) \Big|_{x=\frac{\pi}{3}} = \frac{\sqrt{3}}{2}$

LIMITS BASED ON EXPANSION

27. Let $l_1 = \lim_{x \rightarrow 0} \frac{\sin(\sin^{-1}x) - \sin x}{x^3}$, $l_2 = \lim_{x \rightarrow 0} \frac{\tan(\tan^{-1}x) - \ln(1+x)}{x^2}$, $l_3 = \lim_{x \rightarrow 0} \frac{e^x - e^{\ln x} - 1}{x^2}$ & $l_4 = \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x^3}$
 $l_5 = \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - x^2 - 2}{x^4}$. Find the value of $3(l_1 + 2l_2 + 3l_3 + l_4 \div l_5)$

माना $l_1 = \lim_{x \rightarrow 0} \frac{\sin(\sin^{-1}x) - \sin x}{x^3}$, $l_2 = \lim_{x \rightarrow 0} \frac{\tan(\tan^{-1}x) - \ln(1+x)}{x^2}$, $l_3 = \lim_{x \rightarrow 0} \frac{e^x - e^{\ln x} - 1}{x^2}$ & $l_4 = \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x^3}$
 $l_5 = \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - x^2 - 2}{x^4}$. $3(l_1 + 2l_2 + 3l_3 + l_4 \div l_5)$ का मान ज्ञात करें—

Ans. 20

Sol. $l_1 = \lim_{x \rightarrow 0} \frac{\sin(\sin^{-1}x) - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \frac{1}{6}$

$l_2 = \lim_{x \rightarrow 0} \frac{\tan(\tan^{-1}x) - \ln(1+x)}{x^2} = \lim_{x \rightarrow 0} \frac{x - \ln(1+x)}{x^2} = \frac{1}{2}$

$$I_3 = \lim_{x \rightarrow 0} \frac{e^x - e^{\ln x} - 1}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - x - 1}{x^2} = \frac{1}{2}$$

$$I_4 = \lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x^3} = \lim_{x \rightarrow 0} \frac{\left(1 + x + \frac{x^2}{2} + \frac{x^3}{3} + \dots\right) - \left(1 - x + \frac{x^2}{2} - \frac{x^3}{3} + \dots\right) - 2x}{x^3}$$

$$= \frac{2}{3} = \frac{1}{3}$$

$$I_5 = \lim_{x \rightarrow 0} \frac{e^x + e^{-x} - x^2 - 2}{x^4} = \frac{2}{4} = \frac{1}{12}$$

$$\therefore 3 \left(I_1 + 2I_2 + 3I_3 + \frac{I_4}{I_5} \right) = 20$$

28. $\lim_{x \rightarrow 1} \left(\frac{p}{1-x^p} - \frac{q}{1-x^q} \right)$, $p, q \in \mathbb{N}$

Ans. $\frac{p-q}{2}$

Sol. $\lim_{x \rightarrow 1} \left(\frac{p}{1-x^p} - \frac{1}{1-x^q} \right) = \lim_{x \rightarrow 1} \left(\frac{p - px^q - q + qx^p}{(1-x^p)(1-x^q)} \right)$

$$= \lim_{x \rightarrow 1} \frac{(p-q) + qx^p - px^q}{(1-x^p)(1-x^q)}$$

$$= \lim_{h \rightarrow 0} \frac{(p-q) + q(1+h)^p - p(1+h)^q}{(1-(1+h)^q)(1-(1+h)^p)}$$

$$= \lim_{h \rightarrow 0} \frac{(p-q) + q \left(1 + ph + \frac{p(p-1)h^2}{2} + \dots \right) - p \left(1 + qh + \frac{q(q-1)h^2}{2} + \dots \right)}{\left(1 - \left(1 + qh + \frac{q(q-1)h^2}{2} + \dots \right) \right) \left(1 - \left(1 + ph + \frac{p(p-1)h^2}{2} + \dots \right) \right)}$$

$$= \frac{\frac{pq}{2} - \frac{pq(q-1)}{2}}{pq} = \frac{p-q}{2}$$

29. $\lim_{x \rightarrow \infty} x^2 \left(\sqrt{\frac{x+2}{x}} - \sqrt[3]{\frac{x+3}{x}} \right)$

Ans. 1/2

Sol. $\lim_{x \rightarrow \infty} x^2 \left(\sqrt{\frac{x+2}{x}} - \left(\frac{x+3}{x} \right)^{\frac{1}{3}} \right)$

$$= \lim_{x \rightarrow \infty} \frac{\sqrt{1 + \frac{2}{x}} - \left(1 + \frac{3}{x} \right)^{\frac{1}{3}}}{\frac{1}{x^2}}$$

$$= \lim_{x \rightarrow \infty} \frac{\left(1 + \frac{2}{x} \right)^{\frac{1}{2}} - \left(1 + \frac{3}{x} \right)^{\frac{1}{3}}}{\frac{1}{x^2}}$$

$$= \frac{1 + \frac{1}{2} \left(\frac{2}{x} \right) + \frac{-1}{4} \left(\frac{2}{x} \right)^2 \cdot \frac{1}{2} + \dots - \left(1 + \frac{1}{3} \left(\frac{3}{x} \right) + \frac{1}{3} \left(\frac{3}{x} - 1 \right) \left(\frac{3}{x} \right)^2 \cdot \frac{1}{2} + \dots \right)}{\frac{1}{x^2}}$$

$$= -\frac{1}{2} + 1 = \frac{1}{2}$$

30. Find a & b if: (i) $\lim_{x \rightarrow \infty} \left[\frac{x^2 + 1}{x + 1} - ax - b \right] = 0$ (ii) $\lim_{x \rightarrow -\infty} \left[\sqrt{x^2 - x + 1} - ax - b \right] = 0$

a व b ज्ञात करिये जहाँ पर: (i) $\lim_{x \rightarrow \infty} \left[\frac{x^2 + 1}{x + 1} - ax - b \right] = 0$ (ii) $\lim_{x \rightarrow -\infty} \left[\sqrt{x^2 - x + 1} - ax - b \right] = 0$

Ans. (i) $a=1$, $b=-1$ (ii) $a=-1$, $b=\frac{1}{2}$

Sol. We put $x = -t$

$$\lim_{t \rightarrow \infty} \left(\sqrt{t^2 + t + 1} + at - b \right) = 0$$

[For $\infty - \infty$, a must be negative.]

$$\lim_{t \rightarrow \infty} \left(\frac{t^2 + t + 1 - (at - b)^2}{\sqrt{t^2 + t + 1} - (at - b)} \right) = 0$$

$$\lim_{t \rightarrow 0} \left(\frac{t^2(1-a^2) + t(1+2ab) + 1-b^2}{\sqrt{t^2+t+1} - (at-b)} \right)$$

For limit coefficient of t^2 in numerator should be zero

$$1 - a^2 = 0 \text{ \& } 1 + 2ab = 0$$

$$\Rightarrow a^2 = \pm 1 \Rightarrow a = -1 \Rightarrow b = \frac{1}{2}$$

($a=1$ is rejected)

31. If the $\lim_{x \rightarrow 0} \frac{1}{x^3} \left(\frac{1}{\sqrt{1+x}} - \frac{1+ax}{1+bx} \right)$ exists and has the value equal to l , then find the value of $\frac{1}{a} - \frac{2}{l} + \frac{3}{b}$.

यदि $\lim_{x \rightarrow 0} \frac{1}{x^3} \left(\frac{1}{\sqrt{1+x}} - \frac{1+ax}{1+bx} \right)$ विद्यमान हो और इसका मान l हो तो $\frac{1}{a} - \frac{2}{l} + \frac{3}{b}$ का मान ज्ञात करो।

Ans. 72

Sol.
$$\lim_{x \rightarrow 0} \frac{1}{x^3} \left(\frac{1+bx - (1+ax)\sqrt{1+x}}{\sqrt{1+x}(1+bx)} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^3} \left(\frac{(1+bx)^2 - (1+ax)^2(1+x)}{\sqrt{1+x}(1+bx)(1+bx + (1+ax)\sqrt{1+x})} \right)$$

$$= \lim_{x \rightarrow 0} \frac{1}{x^3} \frac{1+b^2x^2+2bx - (1+a^2x^2+2ax+2ax^2+a^2x^3+x)}{\sqrt{1+x}(1+bx)(1+bx + (1+ax)\sqrt{1+x})}$$

$$= \lim_{x \rightarrow 0} \frac{-a^2x^3 + (b^2 - a^2 - 2a)x^2 + (2b - 2a - 1)x}{2x^3}$$

$$\Rightarrow 1 = -\frac{a^2}{2} \quad \& \quad b^2 - a^2 - 2a = 0 \quad \& \quad 2b - 2a - 1 = 0$$

$$\Rightarrow b^2 = a^2 + 2a \quad \& \quad 2b = 2a + 1$$

$\downarrow$
 $a = \frac{1}{4}, b = \frac{3}{4}$

$$\Rightarrow \frac{1}{a} - \frac{2}{l} + \frac{3}{b} = 4 + \frac{4}{a^2} + \frac{3}{\frac{3}{4}} = 8 + 64 = 72$$

32. Let $f(x) = ax^3 + bx^2 + cx + d$ and $g(x) = x^2 + x - 2$.

If $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = 1$ and $\lim_{x \rightarrow -2} \frac{f(x)}{g(x)} = 4$, then find the value of $\frac{c^2 + d^2}{a^2 + b^2}$.

माना $f(x) = ax^3 + bx^2 + cx + d$ तथा $g(x) = x^2 + x - 2$ है।

यदि $\lim_{x \rightarrow 1} \frac{f(x)}{g(x)} = 1$ तथा $\lim_{x \rightarrow -2} \frac{f(x)}{g(x)} = 4$ है, तो $\frac{c^2 + d^2}{a^2 + b^2}$ का मान ज्ञात कीजिये।

Ans. 16

Sol. $g(x) = x^2 + x - 2 = (x + 2)(x - 1)$

Now $\lim_{x \rightarrow 1} \frac{f(x)}{(x + 2)(x - 1)}$ and $\lim_{x \rightarrow -2} \frac{f(x)}{(x + 2)(x - 1)}$ both exist

hence $f(x) = (x - 1)(x + 2)(k_1x + k_2)$

$$\therefore \lim_{x \rightarrow 1} (k_1x + k_2) = 1 \Rightarrow k_1 + k_2 = 1 \quad \dots(1)$$

$$\text{and } \lim_{x \rightarrow -2} (k_1x + k_2) = 4 \Rightarrow k_2 - 2k_1 = 4 \quad \dots(2)$$

On solving (1) and (2), we get

$$k_1 = -1 \text{ and } k_2 = 2$$

Hence cubic $f(x) = (x - 1)(x + 2)(2 - x) = -x^3 + x^2 + 4x - 4$

$$\therefore a = -1, b = 1, c = 4, d = -4$$

$$\text{Hence } \frac{c^2 + d^2}{a^2 + b^2} = \frac{(4)^2 + (-4)^2}{(-1)^2 + (1)^2} = 16$$

33. Find the values of constants a, b & c so that $\lim_{x \rightarrow 0} \frac{axe^x - b \log(1+x) + cxe^{-x}}{x^2 \sin x} = 2$.

अचर a, b एवं c के मान ज्ञात कीजिए ताकि $\lim_{x \rightarrow 0} \frac{axe^x - b \log(1+x) + cxe^{-x}}{x^2 \sin x} = 2$ हो.

Ans. $a = 3, b = 12, c = 9$

$$\text{Sol. } \lim_{x \rightarrow 0} \frac{ax \left(1 + x + \frac{x^2}{2!} + \dots \right) - b \left(x - \frac{x^2}{2} + \frac{x^3}{3} - \dots \right) + cx \left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots \right)}{x^2 \left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right)} = 2$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{(a - b + c) + x \left(a + \frac{b}{2} - c \right) + x^2 \left(\frac{a}{2} - \frac{b}{3} + \frac{c}{2} \right) + x^3 \text{ and term containing higher power}}{x^2 - \frac{x^4}{3!} + \frac{x^6}{5!} - \dots} = 2$$

$$\text{then } a - b + c = 0 \quad \dots(1)$$

$$2a + b - 2c = 0 \quad \dots(2)$$

$$3a - 2b + 3c = 12 \quad \dots(3)$$

On solving (1), (2), (3)

we get $a = 3, b = 12, c = 9$

34. If $\lim_{x \rightarrow \infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1$, then the value of $(a + b + c)$ can be expressed in the

lowest form as $\frac{p}{q}$. Find the value of $(p + q)$.

यदि $\lim_{x \rightarrow \infty} \frac{a(2x^3 - x^2) + b(x^3 + 5x^2 - 1) - c(3x^3 + x^2)}{a(5x^4 - x) - bx^4 + c(4x^4 + 1) + 2x^2 + 5x} = 1$, तो $(a + b + c)$ का मान $\frac{p}{q}$ के सरलतम रूप में व्यक्त किया जा

सकता है तो $(p + q)$ का मान ज्ञात करें।

Ans. 167

Sol. $\lim_{x \rightarrow \infty} \frac{(2a + b - 3c)x^3 + x^2(5b - c - a) - b}{(5a - b + 4c)x^4 + 2x^2 + (5 - a)x + c} = 1$

$$\Rightarrow \left. \begin{array}{l} 2a + b - 3c = 0 \\ 5a - b + 4c = 0 \\ 5b - c - a = 2 \end{array} \right\} \rightarrow \left. \begin{array}{l} 7a + c = 0 \\ 4a + 3c + 4b = 2 \end{array} \right\} \rightarrow 4(a + b) - 21a = 2$$

$$4b - 17a = 2$$

$$8a + 4b - 12c = 0 \Rightarrow 8a + 4b + 84a = 0$$

$$\Rightarrow 4b = -92a$$

$$\Rightarrow b = -23a$$

$$\Rightarrow a = \frac{-2}{109} \quad \Rightarrow a + (-23a) + -7a = -29a$$

$$= \frac{58}{109}$$

$$\Rightarrow p + q = 167$$

35. $f(x)$ is the function such that $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$. If $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{(f(x))^3} = 1$, then find the value of

a and b .

$f(x)$ एक ऐसा फलन है कि $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$ यदि $\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{(f(x))^3} = 1$, तो a व b का मान ज्ञात करो।

Ans. $a = -5/2, b = -3/2$

Sol. $\lim_{x \rightarrow 0} \frac{f(x)}{x} = 1$

$$\lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{(f(x))^3} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x + ax \left(1 - \frac{x^2}{2} + \frac{x^4}{24} + \dots \right) - b \left(x - \frac{x^3}{6} + \dots \right)}{x^3 \cdot \frac{(f(x))^3}{x^3}} = 1$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x(1 + a - b) + \left(\frac{b}{6} - \frac{a}{2} \right) x^3 + \dots}{x^3} = 1$$

$$\Rightarrow a - b + 1 = 0 \quad \& \quad \frac{b}{6} - \frac{a}{2} = 1$$

$$\Rightarrow \underbrace{a - b + 1 = 0 \quad \& \quad b - 3a = 6}_{\downarrow}$$

$$\Rightarrow \begin{aligned} a - (6 + 3a) + 1 &= 0 \\ \Rightarrow -2a - 5 &= 0 \end{aligned}$$

$$a = -\frac{5}{2}$$

$$b = -\frac{15}{2} + 6 = \frac{12 - 15}{2} = -\frac{3}{2}$$

36. If $\lim_{x \rightarrow 1} (1 + ax + bx^2)^{\frac{c}{x-1}} = e^3$, then the value of $\left| \frac{5b}{a} \right| + 4bc$ is equal to.

यदि $\lim_{x \rightarrow 1} (1 + ax + bx^2)^{\frac{c}{x-1}} = e^3$, तो $\left| \frac{5b}{a} \right| + 4bc$ का मान बराबर होगा।

Ans. 17

Sol. $\lim_{x \rightarrow 1} (1 + ax + bx^2)^{\frac{c}{x-1}} = e^3$

$$\Rightarrow 1 + a + b = 1$$

$$\Rightarrow \exp\left(\lim_{x \rightarrow 1} \frac{acx + bcx^2}{x-1}\right) = e^3$$

$$\Rightarrow (a+b)c = 0$$

$$\Rightarrow \lim_{x \rightarrow 1} c \left(\frac{ax + bx^2}{x-1} \right) = 3$$

$$\Rightarrow c \lim_{x \rightarrow 1} \frac{a(x-1+1) + b(x-1)(x+1) + b}{x-1} = 3$$

$$\Rightarrow c \lim_{x \rightarrow 1} \left(a + b(x+1) + \frac{a+b}{x-1} \right) = 3$$

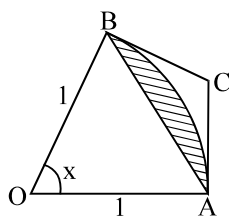
$$\Rightarrow (a+2b)c = 3$$

$$\Rightarrow bc = 3$$

$$\Rightarrow \left| \frac{5b}{a} \right| + 4bc = 17$$

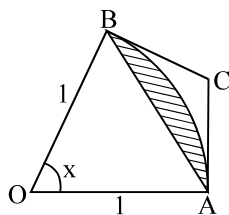
GEOMETRICAL LIMITS

37. A circular arc of radius 1 subtends an angle of x radians, $0 < x < \frac{\pi}{2}$ as shown in the figure. The point C is the intersection of the two tangent lines at A & B. Let $T(x)$ be the area of triangle ABC & let $S(x)$ be the area of the shaded region. Compute:



- (a) $T(x)$ (b) $S(x)$ (c) the limit of $\frac{T(x)}{S(x)}$ as $x \rightarrow 0$.

त्रिज्या 1 का एक वृत्ताकार चाप चित्रानुसार x रेडियन का कोण बनाता है, जहाँ $0 < x < \frac{\pi}{2}$. बिन्दु A व B पर खींची गयी स्पर्श रेखा बिन्दु C पर प्रतिच्छेद करती है। यदि $T(x)$ त्रिभुज ABC का क्षेत्रफल तथा $S(x)$ रेखांकित क्षेत्र का क्षेत्रफल है तो निम्नलिखित के मान की गणना करो।



(a) $T(x)$ (b) $S(x)$ (c) $\frac{T(x)}{S(x)}$ की $x \rightarrow 0$ पर सीमा

Ans. $T(x) = \frac{1}{2} \tan^2 \frac{x}{2} \cdot \sin x$ or $\tan \frac{x}{2} - \frac{\sin x}{2}$, $S(x) = \frac{1}{2} x - \frac{1}{2} \sin x$, limit $= \frac{3}{2}$

Sol. $T(x) = \frac{1}{2} BC \cdot AC \sin(\pi - x)$

$$= \frac{1}{2} \left(\tan^2 \frac{x}{2} \right) \cdot \sin x = \tan \frac{x}{2} - \frac{\sin x}{2}$$

$s(x) = \text{area of sector OAB} - \text{area of } \Delta OAB$

$$= \frac{1}{2} (1)^2 \cdot x - \frac{1}{2} (1)^2 \sin x$$

$$\lim_{x \rightarrow 0} \frac{\tan \frac{x}{2} - \frac{\sin x}{2}}{\frac{x - \sin x}{2}} = \frac{3}{2}$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 38 to 40

Consider two functions $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n$ and $g(x) = -x^{4b}$ where $b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right)$. Then :

माना दो फलन $f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n$ तथा $g(x) = -x^{4b}$ है, जहाँ $b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right)$. तो :

38. $f(x)$ is :

$f(x)$ है :

- (A) e^{-x^2} (B) $e^{\frac{-x^2}{2}}$ (C) e^{x^2} (D) $e^{\frac{x^2}{2}}$

Ans. B

39. $g(x)$ is
 $g(x) =$
 (A) $-x^2$ (B) x^2 (C) x^4 (D) $-x^4$

Ans. A

40. Number of solutions of $f(x) + g(x) = 0$ is :

$f(x) + g(x) = 0$ के हलों की संख्या है -

- (A) 2 (B) 4 (C) 0 (D) 1

Ans. A

Sol. (38 - 40)

$$f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n = e^{\lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} - 1 \right) \cdot n}$$

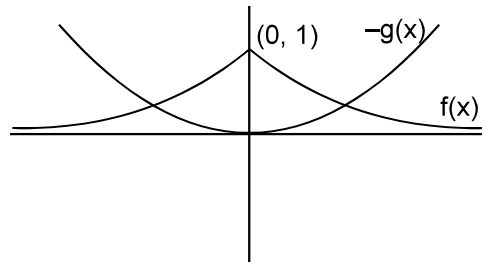
$$\text{Substituting, } n = \frac{1}{t^2}$$

$$f(x) = e^{\lim_{t \rightarrow 0} \left(\frac{\cos tx - 1}{t^2} \right)} = e^{\lim_{t \rightarrow 0} -x^2 \left(\frac{1 - \cos tx}{t^2 x^2} \right)} = e^{-\frac{1}{2}x^2}$$

$$g(x) = -x^{4b}$$

$$b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right) = \lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1 - x^2 - 1}{\sqrt{x^2 + x + 1} + \sqrt{x^2 + 1}} \right) = \frac{1}{2}$$

$$\therefore g(x) = -x^{4 \cdot \frac{1}{2}} = -x^2$$



By observation, graphs of $f(x)$ and $-g(x)$ intersect each other at two points

$\therefore$ Number of solutions is 2.

Hindi. (38-40)

$$f(x) = \lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} \right)^n = e^{\lim_{n \rightarrow \infty} \left(\cos \frac{x}{\sqrt{n}} - 1 \right) \cdot n}$$

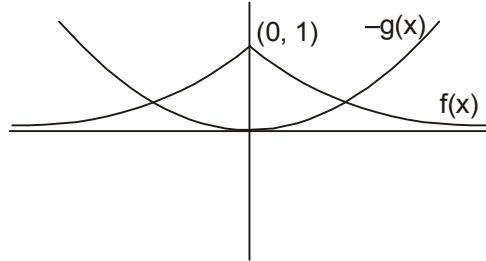
$$n = \frac{1}{t^2} \text{ प्रतिस्थापित करने पर}$$

$$f(x) = e^{\lim_{t \rightarrow 0} \left(\frac{\cos tx - 1}{t^2} \right)} = e^{\lim_{t \rightarrow 0} -x^2 \left(\frac{1 - \cos tx}{t^2 x^2} \right)} = e^{-\frac{1}{2}x^2}$$

$$g(x) = -x^{4b}$$

$$b = \lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 + 1} \right) = \lim_{x \rightarrow \infty} \left(\frac{x^2 + x + 1 - x^2 - 1}{\sqrt{x^2 + x + 1} + \sqrt{x^2 + 1}} \right) = \frac{1}{2}$$

$$\therefore g(x) = -x^{4\frac{1}{2}} = -x^2$$



देखने पर पता चलता है कि आरेख $f(x)$ तथा $-g(x)$ एक दूसरे को दो बिन्दुओं पर प्रतिच्छेद करते हैं।

$\therefore$ हलों की संख्या 2 है।

Paragraph for Question Nos. 41 to 43

Consider $f(x) = \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$, where a, b, c are real numbers.

माना $f(x) = \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$, जहाँ a, b, c वास्तविक संख्याएँ हैं।

41. If $\lim_{x \rightarrow 0^+} f(x)$ is finite, then the value of $a + b + c$ is

यदि $\lim_{x \rightarrow 0^+} f(x)$ परिमित है तो $a + b + c$ का मान होगा –

- (A) 0 (B) 1 (C) 2 (D) -2

Ans. A

42. If $\lim_{x \rightarrow 0^+} f(x) = \ell$ (finite), then the value of ℓ is

यदि $\lim_{x \rightarrow 0^+} f(x) = \ell$ (परिमित), तो ℓ का मान होगा –

- (A) -2 (B) $-\frac{1}{2}$ (C) -1 (D) $-\frac{1}{3}$

Ans. D

43. Using the values of a, b, c as found above, the value of $\lim_{x \rightarrow 0^+} x f(x)$ is :

उपरोक्त ज्ञात किये गये a, b, c के मानों का उपयोग करते हुये $\lim_{x \rightarrow 0^+} x f(x)$ का मान होगा –

- (A) 0 (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) 2

Ans. A

Sol. (41-43)

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^3}$$

$$= \lim_{x \rightarrow 0^+} \frac{\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} + \dots\right) + a\left(1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots\right) + b\left(1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \dots\right) + c\left(x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots\right)}{x^3}$$

$$= \lim_{x \rightarrow 0^+} \frac{(a+b) + x(1+a-b) + x^2\left(\frac{a}{2!} + \frac{b}{2!} - \frac{c}{2}\right) + x^3\left(-\frac{1}{3!} + \frac{a}{3!} - \frac{b}{3!} + \frac{c}{3}\right) + x^4\left(\dots\right) + \dots}{x^3}$$

$$\therefore \text{limit is finite} \Rightarrow \left. \begin{array}{l} a+b=0 \\ 1+a-b=0 \\ a+b-c=0 \end{array} \right\} \Rightarrow a = -\frac{1}{2}, b = \frac{1}{2} \text{ and } c = 0$$

$$\text{Also } \lim_{x \rightarrow 0^+} f(x) = -\frac{1}{3!} + \frac{a}{3!} - \frac{b}{3!} + \frac{c}{3} = -\frac{1}{6}(1-a+b-2c) = -\frac{1}{3}$$

$$4. \quad a + b + c = -\frac{1}{2} + \frac{1}{2} + 0 = 0$$

$$5. \quad \lim_{x \rightarrow 0^+} f(x) = -\frac{1}{3}$$

$$6. \quad \lim_{x \rightarrow 0^+} x f(x) = \lim_{x \rightarrow 0^+} \frac{\sin x + ae^x + be^{-x} + c \ln(1+x)}{x^2} = \frac{a+b-c}{2} = 0$$

Paragraph for Question Nos. 44 to 45

$$\text{Let } f(x) = \begin{cases} x+3 & ; -2 < x < 0 \\ 4 & ; x = 0 \\ 2x+5 & ; 0 < x < 1 \end{cases}, \text{ then :}$$

$$\text{माना } f(x) = \begin{cases} x+3 & ; -2 < x < 0 \\ 4 & ; x = 0 \\ 2x+5 & ; 0 < x < 1 \end{cases}, \text{ है तो}$$

44. $\lim_{x \rightarrow 0^-} f([x - \tan x])$ is : (where $[.]$ denotes greatest integer function)

- (A) 2 (B) 4 (C) 5 (D) None of these

$\lim_{x \rightarrow 0^-} f([x - \tan x])$ का मान होगा : (जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है)

- (A) 2 (B) 4 (C) 5 (D) इनमें से कोई नहीं

Ans. B

45. $\lim_{x \rightarrow 0^+} f\left(\left\{\frac{x}{\tan x}\right\}\right)$ is : (where $\{.\}$ denotes fractional part of function)

- (A) 4 (B) 5 (C) 7 (D) None of these

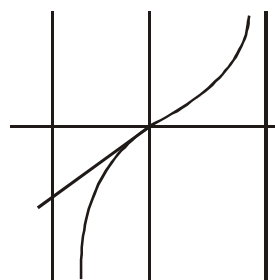
$\lim_{x \rightarrow 0^+} f\left(\left\{\frac{x}{\tan x}\right\}\right)$ का मान होगा : (जहाँ $\{.\}$ भिन्नात्मक भाग फलन को दर्शाता है)

- (A) 4 (B) 5 (C) 7 (D) कोई नहीं

Ans. C

Sol. (44 – 45)

$$f(x) = \begin{cases} x+3 & -2 < x < 0 \\ 4 & x = 0 \\ 2x+5 & 0 < x < 1 \end{cases}$$



$$\lim_{x \rightarrow 0^-} f([x - \tan x])$$

$$\because x \geq \tan x$$

$$\Rightarrow x - \tan x \geq 0$$

$$x - \tan x = \begin{cases} < 0 & \text{if } x > 0 \\ 0 & x = 0 \\ > 0 & \text{if } x < 0 \end{cases}$$

$$x - \tan x = \begin{cases} -1 & \text{if } x > 0 \\ 0 & x = 0 \\ 0 & x < 0 \end{cases}$$

$$\Rightarrow \lim_{x \rightarrow 0^-} f(0) = 4$$

$$\lim_{x \rightarrow 0^+} f\left(\left\{\frac{x}{\tan x}\right\}\right) = \lim_{x \rightarrow 0^+} 2\left\{\frac{x}{\tan x}\right\} + 5$$

$$= \lim_{x \rightarrow 0^+} \frac{2x}{\tan x} + 5 = 7$$

ONE OR MORE THAN ONE CORRECT TYPE

46. If $\lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^3} = p$ (finite), then :

यदि $\lim_{x \rightarrow 0} \frac{\sin 2x + a \sin x}{x^3} = p$ (परिमित) हो, तो :

- (A) $a = -2$ (B) $a = -1$ (C) $p = -2$ (D) $p = -1$

Ans. AD

Sol. $\lim_{x \rightarrow 0} \frac{2 \sin x \cos x + a \sin x}{x^3} = p$ (finite परिमित)

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \left(\frac{2 \cos x + a}{x^2} \right) = p \Rightarrow \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \lim_{x \rightarrow 0} \frac{2 \cos x + a}{x^2} = p$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{2 \cos x + a}{x^2} = p$$

For $\left(\frac{0}{0} \right)$ form $\left(\frac{0}{0} \right)$ रूप के लिए

$$2 + a = 0$$

$$a = -2$$

$$\Rightarrow p = \lim_{x \rightarrow 0} \frac{2(\cos x - 1)}{x^2} \Rightarrow p = \lim_{x \rightarrow 0} \frac{-2 \left(2 \sin^2 \frac{x}{2} \right)}{x^2} \Rightarrow p = -1$$

47. Let $f(x) = \frac{|x + \pi|}{\sin x}$, then

- (A) $f(-\pi^+) = -1$ (B) $f(-\pi^-) = 1$
(C) $\lim_{x \rightarrow -\pi} f(x)$ does not exist (D) $\lim_{x \rightarrow \pi} f(x)$ does not exist

माना $f(x) = \frac{|x + \pi|}{\sin x}$ हो, तो—

- (A) $f(-\pi^+) = -1$ (B) $f(-\pi^-) = 1$
(C) $\lim_{x \rightarrow -\pi} f(x)$ विद्यमान नहीं है। (D) $\lim_{x \rightarrow \pi} f(x)$ विद्यमान नहीं है।

Ans. ABCD

Sol. $f(x) = \frac{|x + \pi|}{\sin x}$

$$(A) f(-\pi^+) = \lim_{h \rightarrow 0} \frac{|-\pi + h + \pi|}{\sin(-\pi + h)} = \lim_{h \rightarrow 0} \frac{|h|}{-\sin h} = -1$$

$$(B) f(-\pi^-) = \lim_{h \rightarrow 0} \frac{|-\pi - h + \pi|}{\sin(-\pi - h)} = \lim_{h \rightarrow 0} \frac{|h|}{\sin h} = 1$$

(C) $f(-\pi^+) \neq f(-\pi^-)$ so $\lim_{x \rightarrow -\pi} f(x)$ does not exist

(D) for $\lim_{x \rightarrow \pi} f(x)$

$$\text{LHL} = \lim_{x \rightarrow \pi^-} \frac{|x + \pi|}{\sin x} = \lim_{h \rightarrow 0} \frac{2\pi - h}{\sinh} = \frac{2\pi}{0} = \infty$$

$$\text{RHL} = \lim_{x \rightarrow \pi^+} \frac{|x + \pi|}{\sin x} = \lim_{h \rightarrow 0} \frac{2\pi + h}{-\sinh} = -\frac{2\pi}{0} = -\infty$$

$\text{LHL} \neq \text{RHL}$ so $\lim_{x \rightarrow \pi} f(x)$ does not exist.

Hindi. $f(x) = \frac{|x + \pi|}{\sin x}$

(A) $f(-\pi^+) = \lim_{h \rightarrow 0} \frac{|-\pi + h + \pi|}{\sin(-\pi + h)} = \lim_{h \rightarrow 0} \frac{|h|}{-\sinh} = -1$

(B) $f(-\pi^-) = \lim_{h \rightarrow 0} \frac{|-\pi - h + \pi|}{\sin(-\pi - h)} = \lim_{h \rightarrow 0} \frac{|h|}{\sinh} = 1$

(C) $f(-\pi^+) \neq f(-\pi^-)$

अतः $\lim_{x \rightarrow -\pi} f(x)$ विद्यमान नहीं है।

(D) $\lim_{x \rightarrow \pi} f(x)$ के लिये

$$\text{LHL} = \lim_{x \rightarrow \pi^-} \frac{|x + \pi|}{\sin x} = \lim_{h \rightarrow 0} \frac{2\pi - h}{\sinh} = \frac{2\pi}{0} = \infty$$

$$\text{RHL} = \lim_{x \rightarrow \pi^+} \frac{|x + \pi|}{\sin x} = \lim_{h \rightarrow 0} \frac{2\pi + h}{-\sinh} = -\frac{2\pi}{0} = -\infty$$

$\text{LHL} \neq \text{RHL}$

अतः $\lim_{x \rightarrow \pi} f(x)$ विद्यमान नहीं है।

48. $\lim_{x \rightarrow \infty} \frac{(ax + 1)^n}{x^n + A} =$

(A) a^n if $n \in \mathbb{N}$

(B) ∞ if $n \in \mathbb{Z}^-$ & $a = A = 0$

(C) $\frac{1}{1+A}$ if $n = 0$

(D) a^n if $n \in \mathbb{Z}^-, A = 0$ & $a \neq 0$

$$\lim_{x \rightarrow \infty} \frac{(ax + 1)^n}{x^n + A} =$$

(A) a^n यदि $n \in \mathbb{N}$

(B) ∞ यदि $n \in \mathbb{Z}^-$ तथा $a = A = 0$

(C) $\frac{1}{1+A}$ यदि $n = 0$

(D) a^n यदि $n \in \mathbb{Z}^-, A = 0$ तथा $a \neq 0$

Ans. ABCD

Sol. $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A}$

(A) If $n \in \mathbb{N} \Rightarrow \lim_{x \rightarrow \infty} \frac{\left(a + \frac{1}{x}\right)^n}{1 + \frac{A}{x^n}} = \frac{(a+0)^n}{1+0} = a^n$

(B) If $n \in \mathbb{Z}^-$ & $a = A = 0$ then $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{1}{x^n} = \infty \quad n \in \mathbb{Z}^-$

(C) If $n = 0$ then $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{1}{1+A} = \frac{1}{1+A}$

(D) If $n \in \mathbb{Z}^-, A = 0$ & $a \neq 0$ then $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n} = \lim_{x \rightarrow \infty} \left(a + \frac{1}{x}\right)^n = (a+0)^n = a^n$

Hindi. $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A}$

(A) यदि $n \in \mathbb{N} \Rightarrow \lim_{x \rightarrow \infty} \frac{\left(a + \frac{1}{x}\right)^n}{1 + \frac{A}{x^n}} = \frac{(a+0)^n}{1+0} = a^n$

(B) यदि $n \in \mathbb{Z}^-$ & $a = A = 0$

तब $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{1}{x^n} = \infty \quad n \in \mathbb{Z}^-$

(C) यदि $n = 0$

तब $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{1}{1+A} = \frac{1}{1+A}$

(D) यदि $n \in \mathbb{Z}^-, A = 0$ & $a \neq 0$

तब $\lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n + A} = \lim_{x \rightarrow \infty} \frac{(ax+1)^n}{x^n} = \lim_{x \rightarrow \infty} \left(a + \frac{1}{x}\right)^n = (a+0)^n = a^n$

49. Let $f(x) = \frac{\sqrt{x^2+2}}{3x-6} =$

माना $f(x) = \frac{\sqrt{x^2+2}}{3x-6}$ हो, तो—

$$(A) \lim_{x \rightarrow -\infty} f(x) = -\frac{1}{3}$$

$$(B) \lim_{x \rightarrow \infty} f(x) = \frac{1}{3}$$

$$(C) \lim_{x \rightarrow -\infty} f(x) = \frac{1}{3}$$

$$(D) \lim_{x \rightarrow \infty} f(x) = -\frac{1}{3}$$

Ans. AB

Sol.
$$f(x) = \frac{\sqrt{x^2 + 2}}{3x - 2} = \frac{|x| \sqrt{1 + \frac{2}{x}}}{x \left(3 - \frac{2}{x}\right)}$$

$$\Rightarrow f(x) = \frac{|x| \sqrt{1 + \frac{2}{x}}}{x \left(3 - \frac{2}{x}\right)}$$

$$\Rightarrow \lim_{x \rightarrow \infty} f(x) = \frac{1}{3} \quad \& \quad \lim_{x \rightarrow \infty} f(x) = \frac{-1}{3}$$

50. If $\ell = \lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{1 + a \cos x}{x^2} - \lim_{x \rightarrow 0} \frac{b \sin x}{x^3}$, where $\ell \in \mathbb{R}$, then

(A) $(a, b) = (-1, 0)$

(B) a & b are any real numbers

(C) $\ell = 0$

(D) $\ell = \frac{1}{2}$

यदि $\ell = \lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{1 + a \cos x}{x^2} - \lim_{x \rightarrow 0} \frac{b \sin x}{x^3}$, जहाँ $\ell \in \mathbb{R}$ हो, तो –

(A) $(a, b) = (-1, 0)$

(B) a एवं b कोई भी वास्तविक संख्याएँ हैं।

(C) $\ell = 0$

(D) $\ell = \frac{1}{2}$

Ans. AD

Sol.
$$\ell = \lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{x \left(1 + a \left(1 - \frac{x^2}{2} + \frac{x^4}{24} + \dots \right) \right) - b \left(x - \frac{x^3}{6} + \frac{x^5}{120} + \dots \right)}{x^3}$$

$$= \lim_{x \rightarrow 0} \frac{x(1 + a - b) + x^3 \left(\frac{b}{6} - \frac{a}{24} \right) + \dots}{x^3}$$

If ℓ is finite then $\ell = \frac{b}{3} - \frac{a}{2}$ & $a - b + 1 = 0$

If ℓ is not finite then $a - b + 1 \neq 0$

Now, $\ell = \lim_{x \rightarrow 0} \frac{1 + a \cos x}{x^2} - \lim_{x \rightarrow 0} \frac{b \sin x}{x^3} \Rightarrow \ell$ should be finite

$$= \lim_{x \rightarrow 0} \frac{1 + a \left(1 - \frac{x^2}{2} + \frac{x^4}{4} + \dots \right)}{x^2} - \lim_{x \rightarrow 0} \frac{b \left(x - \frac{x^3}{3} + \dots \right)}{x^3}$$

$\downarrow$ $\downarrow$
 should be finite $\Rightarrow 1 + a = 0$ should be finite $\Rightarrow b = 0$
 $\Rightarrow a = -1$ $\Rightarrow \ell = \frac{1}{2}$

51. $\lim_{x \rightarrow c} f(x)$ does not exist when

$\lim_{x \rightarrow c} f(x)$ विद्यमान नहीं है जब

(A) $f(x) = [x] - [2x - 1]$, $c = 3$

(B) $f(x) = [x] - x$, $c = 1$

(C) $f(x) = \{x\}^2 - \{-x\}^2$, $c = 0$

(D) $f(x) = \frac{\tan(\operatorname{sgn} x)}{\operatorname{sgn} x}$, $c = 0$

where $[x]$ denotes step up function & $\{x\}$ fractional part function.

जहाँ $[x]$ महत्तम पूर्णांक फलन को तथा $\{x\}$ भिन्नात्मक भाग फलन को दर्शाता है

Ans. BC

Sol. (A) $\lim_{x \rightarrow 3} [[x] - [2x - 1]]$ exists

as

$$\text{LHL} = \lim_{h \rightarrow 0} [[3 - h] - [6 - 2h - 1]] = 2 - 4 = -2$$

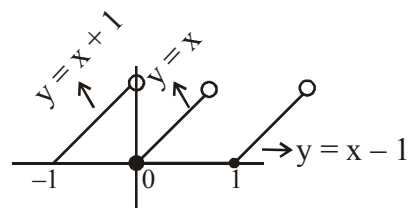
$$\text{RHL} = \lim_{h \rightarrow 0} [[3 + h] - [6 + 2h - 1]] = 3 - 5 = -2$$

(B) $f(x) = [x] - x$

$$\lim_{x \rightarrow 1} f(x) = -\lim_{x \rightarrow 1} \{x\}$$

does not exist

(C) $\lim_{x \rightarrow 0} \{x\}^2 - \{-x\}^2$ does not exist as



$$\text{LHL} = \lim_{h \rightarrow 0} \{-h\}^2 - \{h\}^2 = \lim_{h \rightarrow 0} (-h+1)^2 - h^2$$

$$= 1$$

$$\text{RHL} = \lim_{h \rightarrow 0} \{h\}^2 - \{-h\}^2 = \lim_{h \rightarrow 0} (h^2 - (-h+1)^2)$$

$$(D) \lim_{x \rightarrow 0} \frac{\tan(\operatorname{sgn} x)}{\operatorname{sgn} x} = \tan 1 \text{ exists}$$

52. Which of the following limits vanish ?

निम्न में से किन सीमाओं के मान शून्य है –

$$(A) \lim_{x \rightarrow \infty} x^{\frac{1}{4}} \sin \frac{1}{\sqrt{x}}$$

$$(B) \lim_{x \rightarrow \pi/2} (1 - \sin x) \cdot \tan x$$

$$(C) \lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^2 + x - 5} \cdot \operatorname{sgn}(x)$$

$$(D) \lim_{x \rightarrow 3^+} \frac{[x]^2 - 9}{x^2 - 9}$$

where $[]$ denotes greatest integer function

जहाँ $[]$ महत्तम पूर्णांक फलन है

Ans. ABD

Sol. $\lim_{x \rightarrow 0} \frac{(1 - \sin x) \sin x}{\cos x} \cdot \frac{1 + \sin x}{1 + \sin x}$

$$\lim_{x \rightarrow \pi/2} \frac{\sin x \cos x}{1 + \sin x} = 0]$$

53. If $\lim_{x \rightarrow 0} (p \tan qx^2 - 3 \cos^2 x + 4)^{1/(3x^2)} = e^{5/3}$; $p, q \in \mathbb{R}$ then :

यदि $\lim_{x \rightarrow 0} (p \tan qx^2 - 3 \cos^2 x + 4)^{1/(3x^2)} = e^{5/3}$; $p, q \in \mathbb{R}$ है तो –

$$(A) p = \sqrt{2}, q = \frac{1}{2\sqrt{2}}$$

$$(B) p = \frac{1}{\sqrt{2}}, q = 2\sqrt{2}$$

$$(C) p = 1, q = 2$$

$$(D) p = 2, q = 4$$

Ans. BC

Sol. $\lim_{x \rightarrow 0} (p \tan qx^2 - 3 \cos^2 x + 4)^{\frac{1}{3x^2}} \quad (1^\infty \text{ form})$

$$= \exp \left(\lim_{x \rightarrow 0} \frac{(3 \sin^2 x + p \tan qx^2)}{3x^2} \right)$$

$$= \exp \left(\lim_{x \rightarrow 0} \left(\frac{\sin^2 x}{x^2} + \frac{p}{3} \cdot q \cdot \frac{\tan qx^2}{qx^2} \right) \right)$$

$$= e^{\frac{3+pq}{3}} = e^{\frac{5}{3}}$$

$$\Rightarrow pq = 2$$

54. Let $f: \mathbb{R} \rightarrow [-1, 1]$ be defined as $f(x) = \cos(\sin x)$, then which of the following is/are correct :

(A) f is periodic with fundamental period 2π

(B) Range of $f = [\cos 1, 1]$

(C) $\lim_{x \rightarrow \frac{\pi}{2}} \left(f\left(\frac{\pi}{2} - x\right) + f\left(\frac{\pi}{2} + x\right) \right) = 2$

(D) f is neither even nor odd function

माना $f: \mathbb{R} \rightarrow [-1, 1]$ में फलन $f(x) = \cos(\sin x)$ परिभाषित है तो निम्न में से कौनसा/कौनसे सत्य होंगे –

(A) f एक आवर्ती फलन है जिसका मूलभूत आवर्तकाल 2π है

(B) f का परिसर $[\cos 1, 1]$

(C) $\lim_{x \rightarrow \frac{\pi}{2}} \left(f\left(\frac{\pi}{2} - x\right) + f\left(\frac{\pi}{2} + x\right) \right) = 2$

(D) f ना तो सम और ना ही विषम फलन है

Ans. BC

Sol. $f: \mathbb{R} \rightarrow [-1, 1]$

$$f(x) = \cos(\sin x)$$

(A) $f(x + \pi) = \cos(-\sin x) = f(x)$

(B) $\because -1 \leq \sin x \leq 1$

$$\Rightarrow f(x) \in [\cos 1, 1]$$

(C) $\lim_{x \rightarrow \frac{\pi}{2}} 2 \cos(\cos x) = 2$

(D) $f(-x) = f(x)$

55. If $\lim_{x \rightarrow 0} \frac{(\sin(\sin x) - \sin x)}{ax^3 + bx^5 + c} = -\frac{1}{12}$, then :

यदि $\lim_{x \rightarrow 0} \frac{(\sin(\sin x) - \sin x)}{ax^3 + bx^5 + c} = -\frac{1}{12}$ है तो –

(A) $a = 2$

(B) $a = -2$

(C) $c = 0$

(D) $b \in \mathbb{R}$

Ans. ACD

Sol. $\lim_{x \rightarrow 0} \frac{\sin(\sin x) - \sin x}{ax^3 + bx^5 + c} = -\frac{1}{12}$

$$\Rightarrow c = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin(\sin x) - \sin x}{x^3(a + bx^2)} = -\frac{1}{12}$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\sin x - \frac{\sin^3 x}{3} + \dots - \sin x}{x^3(a + bx^2)} = -\frac{1}{12}$$

$$\Rightarrow \frac{-1}{6a} = \frac{-1}{12}$$

$$\Rightarrow a = 2 \text{ \& } b \in \mathbb{R}$$

MATCH THE COLUMN

56. $\lim_{x \rightarrow 0} f(x)$, where $f(x)$ is as in column-I is :

Column – I

(A) $f(x) = \frac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$

where $[x]$ is the greatest integer function.

(B) $f(x) = \frac{[5/2 + \tan x + \tan^2 x] - [5/2]}{\tan x}$

where $[x]$ is the greatest integer function.

(C) $f(x) = \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$

(D) $f(x) = \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x}$

$\lim_{x \rightarrow 0} f(x)$, जहाँ $f(x)$ स्तम्भ - I में है :

स्तम्भ-I

(A) $f(x) = \frac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$

जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है।

(B) $f(x) = \frac{[5/2 + \tan x + \tan^2 x] - [5/2]}{\tan x}$

जहाँ $[.]$ महत्तम पूर्णांक फलन को दर्शाता है।

Column – II

(P) $\sqrt{2}/8$

(Q) 15

(R) 0

(S) 1/2

स्तम्भ-II

(P) $\sqrt{2}/8$

(Q) 15

$$(C) \quad f(x) = \frac{\sqrt[3]{1+x^2} - \sqrt[4]{1-2x}}{x+x^2}$$

$$(R) \quad 0$$

$$(D) \quad f(x) = \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x}$$

$$(S) \quad 1/2$$

Ans. (A) Q; (B) R; (C) S; (D) P

Sol. (A) $\lim_{x \rightarrow 0} \frac{\tan[e^2]x^2 - \tan[-e^2]x^2}{\sin^2 x}$

$$= \lim_{x \rightarrow 0} \frac{\tan 7x^2 + \tan 8x^2}{\sin^2 x}$$

$$= \lim_{x \rightarrow 0} \frac{\tan 7x^2}{x^2} + \frac{\tan 8x^2}{x^2} = 15$$

$$(B) \quad \lim_{x \rightarrow 0} \frac{\left[\frac{5}{2} + \tan x + \tan^2 x\right] - \left[\frac{5}{2}\right]}{\tan x} = 0$$

$$\because \tan^2 x + \tan x + \frac{5}{2} \in \left[\frac{9}{4}, \infty\right)$$

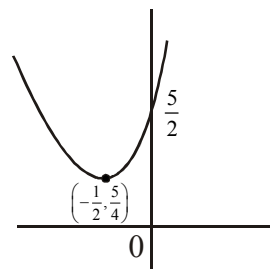
$$\Rightarrow \left[\frac{5}{2} + \tan x + \tan^2 x\right] \in [2, \infty)$$

$$(C) \quad \lim_{x \rightarrow 0} \frac{(1+x^2)^{\frac{1}{3}} - (1-2x)^{\frac{1}{4}}}{x+x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 + \frac{1}{3}(x^2) + \frac{1}{3}\left(\frac{1}{3}-1\right)\frac{x^4}{2} + \dots - \left(1 - \frac{1}{4}(2x) + \frac{1}{4}\left(\frac{1}{4}-1\right)\frac{(2x)^2}{2} + \dots\right)}{x(1+x)}$$

$$= +\frac{1}{2}$$

$$(D) \quad \lim_{x \rightarrow 0} \frac{\sqrt{2} - \sqrt{1+\cos x}}{\sin^2 x} = \lim_{x \rightarrow 0} \frac{\sqrt{2}\left(1 - \cos \frac{x}{2}\right)}{\sin^2 x}$$



$$\begin{aligned}
&= \lim_{x \rightarrow 0} \frac{\sqrt{2} \cdot 2 \sin^2 \frac{x}{4}}{\sin^2 x} \\
&= 2\sqrt{2} \lim_{x \rightarrow 0} \frac{\sin^2 \frac{x}{4}}{\left(\frac{x}{4}\right)^2} \cdot \frac{x^2}{16 \sin^2 x} \\
&= \frac{\sqrt{2}}{8}
\end{aligned}$$

57. Match the following column :

Column-I	Column-II
(A) $\lim_{x \rightarrow \infty} \left(\sqrt{x + \sqrt{x}} - \sqrt{x - \sqrt{x}} \right)$ equals	(P) - 2
(B) The value of the limit, $\lim_{x \rightarrow 0} \frac{\sin 2x - 2 \tan x}{\ln(1 + x^3)}$ is	(Q) - 1
(C) $\lim_{x \rightarrow 0^+} \left(\ln \sin^3 x - \ln(x^4 + e x^3) \right)$ equals	(R) 0
(D) Let $\tan(2\pi \sin \theta) = \cot(2\pi \cos \theta)$, where $\theta \in \mathbb{R}$	(S) 1
and $f(x) = (\sin \theta + \cos \theta)^x$. The value of $\lim_{x \rightarrow \infty} \left[\frac{2}{f(x)} \right]$ equals (Here [] represents greatest integer function)	

निम्न कॉलम को सुमेलित कीजिए :

Column-I	Column-II
(A) $\lim_{x \rightarrow \infty} \left(\sqrt{x + \sqrt{x}} - \sqrt{x - \sqrt{x}} \right)$ का मान होगा	(P) - 2
(B) सीमा $\lim_{x \rightarrow 0} \frac{\sin 2x - 2 \tan x}{\ln(1 + x^3)}$ का मान होगा	(Q) - 1
(C) $\lim_{x \rightarrow 0^+} \left(\ln \sin^3 x - \ln(x^4 + e x^3) \right)$ का मान होगा	(R) 0
(D) माना कि $\tan(2\pi \sin \theta) = \cot(2\pi \cos \theta)$, जहाँ $\theta \in \mathbb{R}$	(S) 1

तथा $f(x) = (|\sin \theta| + |\cos \theta|)^x$ तो $\lim_{x \rightarrow \infty} \left[\frac{2}{f(x)} \right]$ का मान होगा
(जहाँ [] महत्तम पूर्णांक फलन है)

Ans. (A) S; (B) P; (C) Q; (D) R

Sol. (A) $\lim_{x \rightarrow \infty} \frac{(x + \sqrt{x}) - (x - \sqrt{x})}{\sqrt{x + \sqrt{x}} + \sqrt{x - \sqrt{x}}} = \lim_{x \rightarrow \infty} \frac{2\sqrt{x}}{\sqrt{x + \sqrt{x}} + \sqrt{x - \sqrt{x}}} = 1$ Ans.

(C) $\lim_{x \rightarrow 0^+} (\ln \sin^3 x - \ln(x^4 + ex^3)) = \lim_{x \rightarrow 0} \ln \left(\frac{\sin^3 x}{x^3(x + e)} \right) = \ln \left(\frac{1}{e} \right) = -1$

(D) $\tan(2\pi | \sin \theta |) = \tan \left(\frac{\pi}{2} - | \cos \theta | 2\pi \right)$

$$2\pi | \sin \theta | = n\pi + \frac{\pi}{2} - | \cos \theta | 2\pi$$

$$2\pi (| \sin \theta | + | \cos \theta |) = n\pi + \frac{\pi}{2}$$

$$| \sin \theta | + | \cos \theta | = \frac{n}{2} + \frac{1}{4} \quad \dots(1)$$

since $1 \leq | \sin \theta | + | \cos \theta | \leq \sqrt{2}$; $1 \leq \frac{n}{2} + \frac{1}{4} \leq \sqrt{2}$

$$4 \leq 2n + 1 \leq 4\sqrt{2}$$

$$\frac{3}{2} \leq n \leq \frac{4\sqrt{2} - 1}{2}; \text{ Thus } n = 2 \text{ is only possible value.}$$

putting in (i) $| \sin \theta | + | \cos \theta | = \frac{5}{4}$

58. Match the following column :

निम्न कॉलम को सुमेलित कीजिए :

Column-I	Column-II
(A) $\lim_{x \rightarrow 1} \frac{\ln x}{x^4 - 1}$	(P) 2
(B) $\lim_{x \rightarrow 0} \frac{3e^x - x^3 - 3x - 3}{\tan^2 x}$	(Q) $\frac{2}{3}$
(C) $\lim_{x \rightarrow \infty} \frac{\pi - 2 \tan^{-1} x}{\ln \left(1 + \frac{1}{x} \right)}$	(R) $\frac{3}{2}$
(D) $\lim_{x \rightarrow 0} \frac{2 \sin x - \sin 2x}{x(\cos x - \cos 2x)}$	(S) $\frac{1}{4}$
(E) $\lim_{x \rightarrow 0} \frac{e^x - e^{-x} - 2x}{x - \sin x}$	

Ans. (A) S; (B) R; (C) P; (D) Q; (E) P

Sol. (A) $\lim_{x \rightarrow 1} \frac{\ln x}{x^4 - 1} = \lim_{x \rightarrow 1} \frac{\ln(1 + (x-1))}{(x-1)(x^3 + x^2 + x + 1)}$

$$= \frac{1}{4}$$

(B) $\lim_{x \rightarrow 0} \frac{3e^x - x^3 - 3x - 3}{\tan^2 x} = \lim_{x \rightarrow 0} \frac{3e^x - x^3 - 3x - 3}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{3 \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \dots \right) - x^3 - 3x - 3}{x^2}$$

$$= \frac{3}{2}$$

(C) $\lim_{x \rightarrow \infty} \frac{\pi - 2 \tan^{-1} x}{\ln \left(1 + \frac{1}{x} \right)} = \lim_{x \rightarrow \infty} \frac{\pi - 2 \tan^{-1} x}{\frac{1}{x}}$

$$= \lim_{y \rightarrow 0} \frac{\pi - 2 \tan^{-1} \left(\frac{1}{y} \right)}{y}$$

(D) $\lim_{x \rightarrow 0} \frac{2 \sin x (1 - \cos x)}{x 2 \sin \left(\frac{3x}{2} \right) \sin \frac{x}{2}}$

$$= 2 \lim_{x \rightarrow 0} \frac{2 \sin^2 \frac{x}{2}}{2 \sin \frac{x}{2} \cdot \sin \frac{3x}{2}} = \frac{2}{3} \lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{\frac{x}{2}} \cdot \frac{\frac{3x}{2}}{\sin \frac{3x}{2}}$$

$$= \frac{2}{3}$$

(E) $\lim_{x \rightarrow 0} \frac{e^{2x} - 1 - 2xe^x}{e^x (x - \sin x)}$

$$= \lim_{x \rightarrow 0} \frac{\left(1 + 2x + \frac{(2x)^2}{2} + \dots\right) - 1 - 2x \left(1 + x \frac{x^2}{2} + \dots\right)}{\left(x - \left(x - \frac{x^3}{3} + \dots\right)\right)}$$

$$= \lim_{x \rightarrow 0} \frac{x^3 \left(\frac{8}{6} - 1\right)}{x^3 \left(\frac{1}{3} + \dots\right)} = \frac{2}{6} \times 6 = 2$$